

Chapter I

The Problem and Its Background

This chapter serves as the foundation upon which the entire study is built. It introduces the readers to the situation that led to the development of the project and explains why the study was conducted in the first place. This section establishes the context by describing the current situation, identifying the gaps or challenges in existing systems, and presenting the real-world issues that the project seeks to address.

Introduction

Accurate, timely, and comprehensive household data are essential for evidence-based local governance, effective public service delivery, and targeted social protection. At the barangay level—the smallest political and administrative unit in the Philippines—household information underpins activities ranging from health and social welfare interventions to disaster response and local development planning. Yet many barangays still rely on paper-based record keeping or fragmented databases that are prone to data loss, inconsistencies, slow retrieval, and limited analytical capability (Philippine Statistics Authority [PSA], 2024). These gaps constrain local authorities' ability to identify vulnerable households, allocate resources optimally, and monitor the impact of interventions.

Household profiling is the systematic process of enumerating every household within a community and recording the demographic, socio-economic, and service-access characteristics of its members. When properly implemented, household profiling produces granular data that inform targeting for health programs, conditional cash transfers,

livelihood support, and emergency relief operations (Acta Medica Philippina, 2018). Moreover, aggregated profile datasets can reveal population trends, housing conditions, service coverage, and socio-economic disparities—information critical for planning and evidence-driven policymaking (Ojo, 2013). The Philippine government and research institutions have increasingly recognized the value of household and community profiling; recent national efforts such as the PSA’s Household and Community Profiling Survey and the DILG’s Barangay Information Management System (BIMS) reflect a policy direction toward digitized local data systems (PSA, 2024; Department of the Interior and Local Government, 2024).

Despite national initiatives, barangay-level adoption of digital profiling systems remains uneven. Multiple case studies and capstone projects across the country document the benefits of barangay profiling systems—faster record retrieval, automated reporting, improved accuracy, and enhanced decision support—while also noting recurring challenges such as limited local technical capacity, inconsistent data standards, and sustainability constraints (Jacobe et al., 2021; ResearchGate projects on barangay profiling). In many rural and peri-urban barangays, manual registries remain vulnerable to calamities, deterioration, and human error; these conditions weaken the barangay’s capacity to respond rapidly in crises and to implement programs based on reliable beneficiary lists.

Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija is proposed to address these challenges by providing an integrated, secure, and user-centered digital platform for household profiling tailored to the operational context of San Isidro. The system is designed to capture key socio-demographic variables, household

composition, housing characteristics, access to basic services, and program eligibility markers; it offers functionality for registration, updating, search, automated reporting, backup, and role-based access control. By aligning system requirements with international software quality considerations (usability, reliability, security, maintainability, and portability), Barangay-Connect aims to deliver not only functional effectiveness but also long-term sustainability and replicability.

The rationale for Barangay-Connect rests on three interlinked premises. First, digitization of household records reduces the risk of physical loss and accelerates data retrieval, thereby improving the responsiveness of barangay services and emergency interventions (BRISS/BIPS case studies; BARS solutions). Second, standardized and analyzable household data enable evidence-based targeting and monitoring of local programs, thereby increasing the efficiency and equity of resource allocation. Third, implementing a locally contextualized profiling system—developed iteratively with stakeholder participation—promotes user acceptance, data ownership, and operational continuity (e-government literature on participatory system design; Bangero, 2025).

This capstone project adopts the Software Development Life Cycle (SDLC) with an agile, iterative approach to system design and implementation. The research objectives are to: (1) design and develop Barangay-Connect customized to San Isidro’s operational needs; (2) evaluate the system’s technical quality using recognized software quality measures; (3) assess user acceptance and perceived utility among barangay staff and household respondents; and (4) propose recommendations for sustained deployment, maintenance, and potential scaling to other barangays. Evaluation will draw on ISO/IEC

software quality principles and mixed-methods feedback from IT experts, barangay personnel, and resident users.

Barangay-Connect seeks to contribute a practical, evidence-based solution to a pressing local governance need—improving household data integrity and use at the grassroots level. If successfully implemented and institutionalized, the system can enhance San Isidro’s capacity for targeted service delivery, strengthen disaster preparedness, and provide a replicable model for other barangays seeking to transition from fragile manual records to resilient, data-driven local governance.

Review of Related Literature

Efficient data management are essential for barangays in the Philippines, where manual record-keeping presents challenges in accuracy, retrieval, and administrative workload. The development of digital solutions for barangay governance has been explored in several studies, highlighting the advantages of electronic records in local government and public services. This review examines existing research on barangay profiling systems identifying gaps in the literature and justifying the need for Barangay-Connect, a Barangay Household Profiling System designed to streamline household data management and service accessibility.

Development and evaluation of Web-Based Barangay Profiling and Issuance System (BPIS)

The BPIS demonstrates a strong capacity to analyze and predict outcomes within the given dataset, with high correlation and relatively low prediction errors. It provides a clear structure for decision-making and pattern identification. This evaluation is based on

the analysis of 52 instances, with two instances being ignored due to class unknown conditions.

The Computerized Barangay Information System.

According to Martinez (2018), implementing a computerized Management Information System (MIS) for Barangay Calahan would significantly enhance operational efficiency by automating manual processes such as record-keeping and certificate issuance. By adopting an MIS, the barangay can improve service delivery, reduce human errors, and provide timely and accurate information to residents.

According to Claire Ong Carpio (2020), Computerized Barangay Information System would significantly enhance the management of resident records in Barangay Poblacion, Danao City. By transitioning from manual processes to an automated system, the barangay can achieve more efficient record-keeping, faster retrieval of files, and improved accuracy in generating reports. The system's secure storage and backup features ensure data integrity and protection. Overall, this modernization is expected to streamline administrative tasks, reduce workload, and facilitate better service delivery to the community.

The Barangay Profiling System with Analytics

An on-site interview with concerned barangay officials was conducted to determine the range and nature of duties. This matrix served as guide in developing the system modules and interface design. Further, process diagram showing the communication flow from one official to the other was likewise used in the development of the system. The E-

Barangay is a web-based management system where residents can register, file complaints and send requests online. The system allows the barangay secretary to receive and record residents' registry, document requests, and complaints. While the barangay chairman as the approving authority can approve requests, view/respond to complaints and generate reports even when he is on official travel or out of office. In addition, the system is also capable of sending notifications to residents via SMS either on the status of their requests/complaints or for local announcements on advisories and barangay ordinances. Based from the series of test conducted on the functionality and usability of the system, the performance of the E-Barangay as shown on its overall weighted mean was found to be fully functional and usable. Though, there were marginal errors encountered, nevertheless, through prototype refinement, the same was addressed and fixed as reflected on the results of the second system test.

Use of Barangay Information Profiling System (BIPS).

The system has notably enhanced the efficiency and accuracy of record-keeping, issuance of certificates, permits, and clearances, and the management of household and blotter information. By automating these processes, the system has reduced the time and effort required for data entry and retrieval, leading to faster service delivery for residents. Furthermore, the system's secure and organized storage has mitigated issues related to data loss and unauthorized access. The positive feedback from users underscores the system's effectiveness and reliability in addressing the barangay's administrative challenges. (Lacasandile 2020).

Gaps in the Literature

Despite advancements in barangay digitalization, several gaps remain. First, there is a lack of an integrated platform that combines barangay profiling and public services into a single, unified system. Most existing research focuses on large-scale implementations, such as city-wide or province-wide based systems, leaving limited attention to digitalization efforts at the barangay level. Furthermore, many barangays face challenges in adopting such systems due to limited technical knowledge and resources. These factors highlight the need for a user-friendly and cost-effective solution that can effectively support barangay operations and improve community service delivery.

Review of Related Systems

Barangay Profiling System with Analytics (Barangay Pallua Sur, Tuguegarao City)

The developed system automates the profiling of all residents in the barangay. It stores data electronically, thus, records are more organized, and it provides an access of the information needed by the barangay. Reports needed with statistical analysis can be easily generated.

Barangay Profiling System of the City of Candon

The study concludes that the developed computerized profiling system is user-friendly, effective, and aligns with the needs and expectations of users. The mean acceptability rating was very high (4.29), indicating strong satisfaction among stakeholders. The system has potential to enhance public service delivery, safety, and economic development in Candon City. To sustain success, the authors recommend that

local authorities invest in infrastructure, form dedicated implementation teams, engage the community, and provide training to barangay secretaries.

WEB-BON: Design and Development of a Web-Based Barangay Information & Record Management System

The WEB-BON system successfully automated barangay information and record management, replacing manual processes with a web-based solution. Its development, guided by structured software design and the use of PHP and SQL, improved efficiency, accuracy, and accessibility of barangay records.

Automation of Barangay Information Profiling System (BIPS) for Decision Support

The study concluded that the Information-Based Dashboard successfully automated the Barangay Information Profiling System (BIPS), enhancing the efficiency of data processing and decision-making in local governance. With positive evaluations from IT experts and users, the system demonstrated its readiness for implementation, supporting evidence-based planning and resource allocation in the barangay.

Mobile-based Household Profiling System for a Barangay in Quezon, Philippines

A Mobile-based Household Profiling System for Barangay Guinhawa, Quezon, Quezon was designed and developed using Android Studio for mobile development/Android debug, Keystore, simulation, and JetBrains PHP storm. These served as the front-end and back-end of the functionalities of the application. The developed system was highly acceptable in terms of the ISO 25010 standards of compatibility,

functionality suitability, maintainability, performance efficiency, portability, reliability, security, and usability. Thus, it was ready for the utilization of the intended end-users

The researchers successfully achieved the objectives after conducting a sequence of tests and analyses to develop the system. Based on the conclusions and findings, the study recommended that the system features should be improved so that the data management and filtering of the data in Barangay Guinhawa, Quezon, would be made better. Also, the administrator should have the ability to generate reports; the system should have a field that could indicate if the household members have been moved to another place; the developed system may be utilized by the research locale after acquiring software patent registration. Moreover, future researchers may conduct supportive studies that would cater to more features in the system. A memorandum of agreement may also be developed to facilitate the utilization of the developed system in the research locale.

E-Barangay: A Framework for a Web-Based System for Local Communities

The study concluded that the E-Barangay framework effectively addresses the need for community-level e-governance by facilitating core functions such as filing complaints, requesting documents, sending suggestions, and posting announcements. Usability evaluations, both objective and subjective, yielded favorable results, indicating that the system meets stakeholder needs and encourages frequent use.

Barangay Information Management System for Barangay Sampaloc 1

The Barangay Information Management System (BIMS) developed for Barangay Sampaloc 1 in Dasmariñas, Cavite, effectively addresses the challenges of manual record-

keeping by providing a centralized platform for managing resident information. The system's user-friendly interface and functionalities, such as resident registration, profile management, and service request handling, enhance operational efficiency and service delivery

Synthesis

While various digital solutions have enhanced administrative efficiency in the Philippines, there remains a critical gap at the barangay level—specifically, the absence of an integrated system that consolidates household data and other services. Current systems often operate in silos, focusing either on governance or other services, but not both. Barangay-Connect emerges as a strategic innovation that addresses this fragmentation by offering a unified, scalable, and user-friendly platform tailored for barangays. It streamlines data management, improves coordination between residents and other units, and enhances access to essential services—marking a significant step toward inclusive and digitally empowered local governance.

Conceptual Framework

The conceptual framework of this study is anchored on the Input–Process–Output (IPO) model, which illustrates the systematic flow of activities undertaken in developing the Barangay-Connect: Barangay Household Profiling System. This framework serves as the foundation for identifying the essential requirements, development procedures, and desired outcomes of the system.

Input. The Input phase involves gathering all essential information and resources that will serve as the foundation for developing the system, including user requirements and needs analysis from barangay officials, staff, and residents regarding issues in household data collection and management; an assessment of existing data management practices to understand how household records are currently stored, processed, and updated; identification of technical resources such as available hardware, software, internet access, and overall IT infrastructure; and the incorporation of theoretical and conceptual bases such as the System Development Life Cycle (SDLC), Database Management Concepts, and Information Systems Theory to guide the system's design and functionality. Altogether, these inputs provide the necessary groundwork to build an efficient, user-friendly, and reliable profiling system.

Process. The Process represents the series of activities involved in designing, developing, and implementing the Barangay-Connect system, following both SDLC and Agile principles. It includes planning, which defines the system's scope; designing, which outlines how the system will function and how its components will be structured; developing, where coding and programming take place; testing, which identifies defects and ensures the system is free from errors; deployment, which installs the system on user machines and confirms its functionality; review, which evaluates the project's performance based on initial objectives; and maintenance and support, which ensure the system continues to operate properly over time.

Output. The Output of this study is the fully functional Barangay-Connect: Barangay Household Profiling System, a digital platform designed to streamline household data management in San Isidro, Nueva Ecija by efficiently storing, organizing, and

updating household profiles; generating accurate and comprehensive reports for planning and decision-making; providing quick access to essential data for barangay officials and authorized personnel; and reducing manual work while minimizing human errors in record handling.

Feedback. The Feedback or Expected Outcomes of the system implementation include improved accuracy, accessibility, and security of household data; enhanced operational efficiency and greater transparency in local governance; strengthened data-driven decision-making for community planning and service delivery; and increased satisfaction among barangay staff and residents resulting from faster and more reliable services.

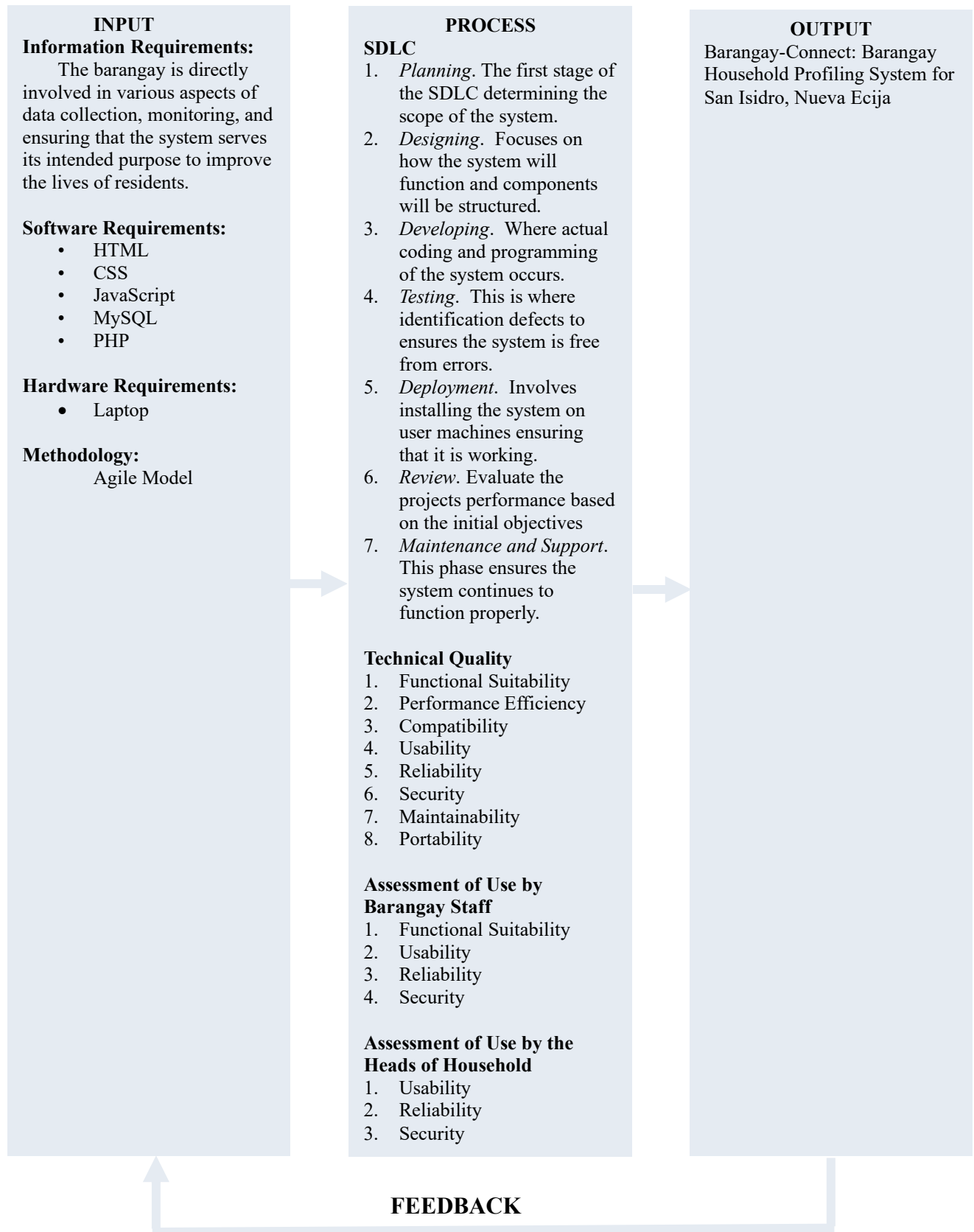


Figure 1. Research Paradigm of Barangay-Connect

Statement of the Problem

The growing demand for efficient resource management, improved social services, and effective community planning has exposed the limitations of traditional household profiling methods, which often rely on paper-based or outdated systems. These methods lead to data inaccuracies, slow service delivery, and difficulty in monitoring community changes. Many organizations and local government units still lack a centralized, real-time household profiling system, making it hard to collect, store, and analyze data efficiently.

As a result, common issues such as human error, poor system integration, limited data accessibility, delayed response times, and ineffective resource allocation persist. To address these problems, this study aims to develop and implement the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija—a centralized and automated platform designed to provide accurate, real-time household information. The system is intended to improve decision-making, streamline service delivery, and support equitable resource distribution for the community. Specifically, this study sought to explain the following:

- 1.1. Planning Phase
- 1.2. Designing Phase
- 1.3. Development Phase
- 1.4. Testing Phase
- 1.5. Deployment Phase
- 1.6. Review Phase
- 1.7. Maintenance and Support Phase

2. Assessment of the technical quality of the Barangay-Connect by the IT Experts based on the International Organization for Standardization (ISO) 25010 Software Quality Standards namely:
 - 2.1. Functional Suitability
 - 2.2. Performance Efficiency
 - 2.3. Compatibility
 - 2.4. Usability
 - 2.5. Reliability
 - 2.6. Security
 - 2.7. Maintainability
 - 2.8. Portability
3. Assessment of Barangay Staff on use of the Barangay-Connect in terms of:
 - 3.1. Functional Suitability
 - 3.2. Usability
 - 3.3. Reliability
 - 3.4. Security
4. Assessment of Head of Households on the use of the Barangay-Connect in terms of:
 - 4.1. Usability
 - 4.2. Compatibility
 - 4.3. Security

Significance of the Study

This study demonstrates the significance of the Barangay-Connect: Barangay Household Profiling System++ by highlighting its potential to streamline barangay governance in San Isidro, Nueva Ecija. By creating a comprehensive and readily accessible household profile database. The result of this study will benefit the following:

To the Barangay Poblacion. The significance of the study of Barangay-Connect for Poblacion Barangay Hall lies in its potential to transform the way barangay officials manage and access resident information. This system is designed to facilitate efficient searching and retrieval of household profiles, particularly by categories (names, age, birthdays, occupation, PWD etc.) which enhances data accessibility and overall administrative efficiency. This enhanced data accessibility directly translates into more efficient transaction processing for residents.

To the Isidoreans. The significance of this study lies in its demonstration of how improved information access enhances the barangay's responsiveness to the needs of its constituents in Isidoreans. Easier access to information leads to faster and simpler transactions, reducing wait times and simplifying interactions. This ultimately improves the overall efficiency and fosters a stronger relationship between the barangay and its people.

To the Students. For students of Bachelor of Science in Information Technology, the project provides practical experience in system development, database design, and application programming. They will also learn how to integrate user interfaces and back-end functionalities in a real-life project. Also, through this study, students can see how to design a system that meets user needs, handles data securely, and ensures smooth

functionality—skills that are directly applicable to IT, software development, and systems analysis careers.

To the Future Researchers. For researchers, the Barangay-Connect: Barangay Household Profiling System++ enables the collection of accurate, comprehensive data that can support evidence-based policy, improve socio-economic research, and contribute to the development of innovative solutions to social and economic challenges. Furthermore, the system fosters interdisciplinary collaboration, enhances the understanding of complex societal issues, and provides a foundation for impactful research that can shape both local and global development efforts.

Scope and Delimitation

The scope of the Household Profiling System project defines its boundaries, key functions, and areas of implementation. The system will initially be deployed in Barangay Poblacion, San Isidro, Nueva Ecija, covering all 1,820 households as the primary test area for evaluating its effectiveness and scalability. It will collect essential demographic information such as household composition, socioeconomic status, and health-related data. A centralized database will store all household profiles, accessible only to authorized local government personnel and integrated with other local databases, particularly health services, for accurate cross-referencing and updates. Key stakeholders include barangay officials responsible for system management, households providing necessary data, and local service providers using the information for resource allocation. Expected outcomes include improved decision-making through accurate, real-time data; more efficient service delivery through streamlined processes; and enhanced community engagement by allowing households to update their own profiles, ensuring data remains current and reliable.

The delimitations of the study define what is excluded from the project, beginning with its limited geographic scope, as it focuses solely on Barangay Poblacion, San Isidro, Nueva Ecija, making the findings less generalizable to other areas. The system will collect only basic demographic, socioeconomic, health, and housing data, excluding more specialized information such as detailed employment histories or medical records unless necessary. Data usage will be restricted to resource allocation and policy planning by local government units, with no application in marketing or commercial activities. The system's performance will depend on the existing network infrastructure, posing challenges for areas with weak connectivity and for households lacking internet access or digital familiarity. Training will be provided before implementation, but post-implementation support will be limited, potentially affecting long-term sustainability. Lastly, the study will be conducted within a 6–12-month timeframe, focusing only on short-term outcomes rather than long-term community or service delivery impacts.

Definition of Terms

Barangay is a type of early Filipino settlement; the word is derived from *balangay*, the name for the sailboats that originally brought settlers of Malay stock to the Philippines from Borneo. Each boat carried a large family group, and the master of the boat retained power as leader, or *datu*, of the village established by his family. (<http://britannica.com/topic/barangay>)

Demographic are statistics that describe populations and their characteristics and usually consists of socioeconomic information expressed statistically including employment, education, income, marriage rates, birth, and death rates, and more.

Household is a social unit consisting of person or a group of persons who sleep in the same housing unit and have a common arrangement in the preparation and consumption of food. (<https://openstat.psa.gov.ph>)

Isidoreans: The residents who live in the town of San Isidro are called Isidoreans.

Profiling is the recording and analysis of a person's psychological and behavioral characteristics, to assess or predict their capabilities in a certain sphere or to assist in identifying a particular subgroup of people. (<https://www.igi-global.com/dictionary/>)

Chapter II

Methodology

This chapter serves as the blueprint or roadmap of the entire study. It explains how the project was carried out, detailing the procedures, techniques, and tools used to design, develop, and evaluate the system or solution. This chapter provides a clear and systematic description of the research or development process, ensuring that the study is transparent, replicable, and scientifically sound.

Research Design

This study adopts a developmental-descriptive research design, a strategy that combines the creation of a technological solution with an empirical investigation of its potential benefits. The primary aim of the research is to design, develop, and assess the Barangay-Connect Household Profiling System—a digital platform intended to improve data management at the barangay level. By integrating descriptive elements, the study not only presents the functionalities of the system but also seeks to understand its perceived impact from the users' perspectives.

In addressing the research questions, this study utilized a mixed-method approach. Quantitative data were gathered through surveys distributed to barangay officials and residents to evaluate the system's usability, efficiency, and relevance. Meanwhile, qualitative data obtained from interviews and observations, provided deeper insights into current practices and the specific challenges faced in barangay profiling and rural health unit coordination.

The selection of respondents follows purposive sampling, targeting individuals who are directly involved in barangay administration and healthcare services. This includes

barangay secretaries, and selected residents who are familiar with the current manual processes. The chosen participants are considered most capable of providing relevant feedback on the system's implementation and usability.

Data were collected through structured questionnaires and guided interviews, which are designed to extract specific information on user experiences, system efficiency, and the current challenges of barangay-level recordkeeping. Observational techniques may also be applied during system demonstrations to identify usability concerns and areas for improvement.

For data analysis, the study used descriptive statistical methods such as frequency distribution, percentages, and weighted mean to interpret quantitative data. For qualitative feedback, thematic analysis was employed to identify common patterns, issues, and suggestions. This dual analysis ensures a comprehensive understanding of both measurable outcomes and subjective experiences.

Overall, the research design supports the objective of creating an innovative system while ensuring that empirical data is used to validate its effectiveness. Through a systematic approach to development, data collection, and analysis, this study aims to contribute a practical, data-driven solution to barangay-level governance and healthcare challenges.

System Development Methodology

The development of Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija followed the Agile Software Development Methodology. The Agile method was chosen because of its flexibility, iterative nature, and strong focus on

user collaboration, which aligns well with the needs of the barangay stakeholders and system developers.

Unlike traditional linear models such as the Waterfall approach, the Agile methodology emphasizes incremental development through continuous feedback, testing, and improvement. This approach allowed the researchers and developers to deliver functional modules of the system in short development cycles known as sprints. After each sprint, the prototype was evaluated and refined based on the feedback from barangay staff.

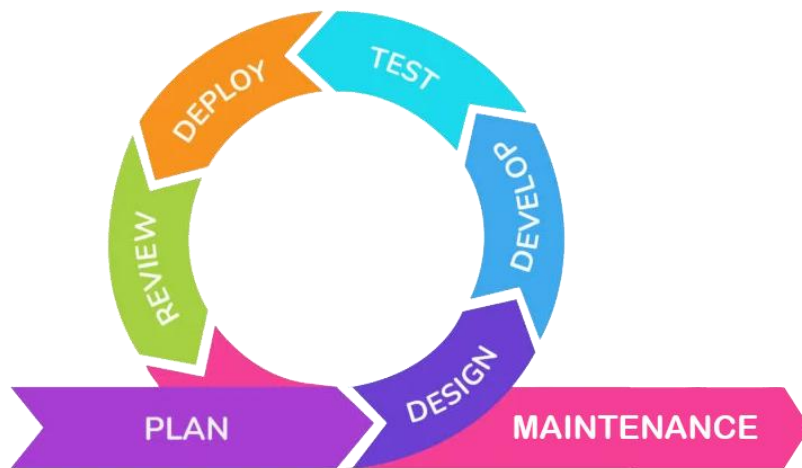


Figure 2. Agile Methodology

Research Locale

This study entitled Barangay-Connect: Barangay Household Profiling System was conducted at the Municipality of San Isidro, Nueva Ecija and piloted at Barangay Poblacion.

The following are some of the reasons why Barangay Poblacion was selected as the research locale for this study:

- a. *Diverse Household Types.* The population includes a range of socioeconomic statuses, from low-income to middle-class, providing a representative sample of community needs.
- b. *Growing Community.* The barangay has seen rapid population growth due to migration from neighboring rural areas, requiring more efficient management of household data.
- c. *Needs for Improved Service Delivery.* The barangay is experiencing challenges with efficiently allocating resources for healthcare, education, and social welfare. With limited data on household needs, resource allocation has been inefficient and reactive, especially in cases of disaster response.
- d. *Interest from Local Government.* The barangay officials have expressed interest in modernizing data collection to improve service delivery and public safety, making it an ideal locale for testing a household profiling system.
- e. *Opportunities for Integration.* The local government is currently developing digital infrastructure and has partnered with the provincial government to implement a more data-driven approach to governance.

By implementing the Barangay-Connect: Barangay Household Profiling System in Barangay Poblacion, this study aims to demonstrate the system's effectiveness in improving local governance, public service delivery, and community engagement in household-level decision-making. The outcomes could provide valuable insights for scaling similar systems to other barangays and municipalities across the Philippines.

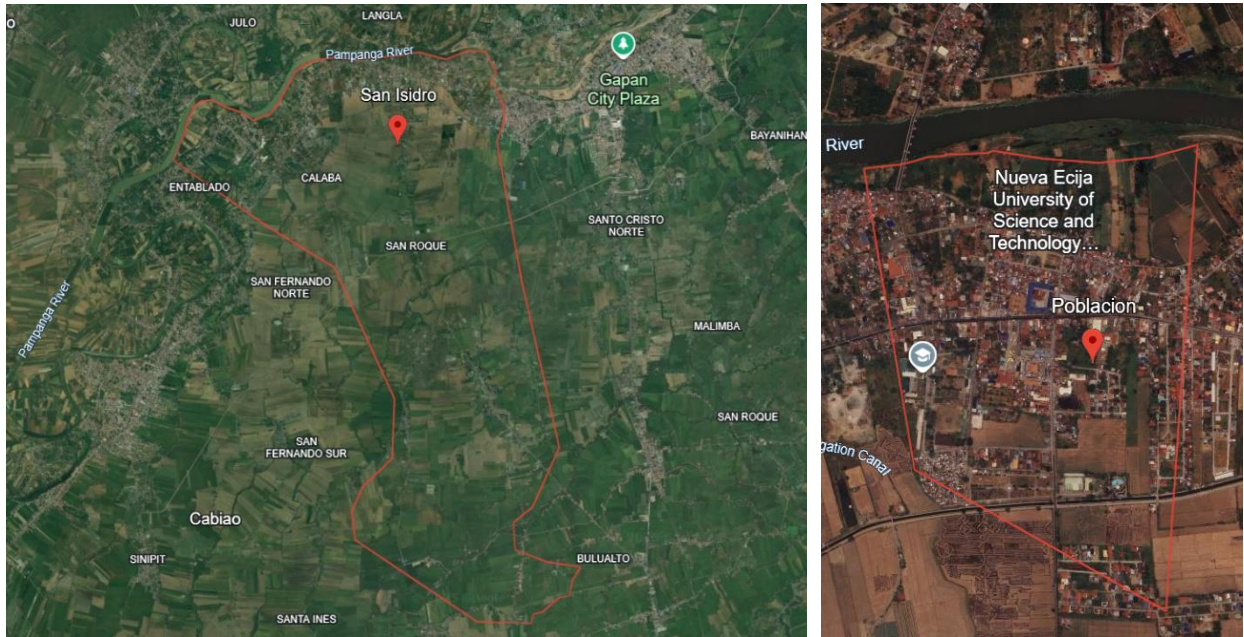


Figure 3. Map of the Municipality of San Isidro, Nueva Ecija (Left) and Barangay Poblacion, San Isidro (Right); the Research Locale

Research Respondents

The respondents of a study focused on a Barangay-Connect: Barangay Household Profiling System typically refer to the individuals or groups who provided the data or information that were analyzed in the system. These respondents play a key role in ensuring the system's data is accurate, reliable, and comprehensive. Below are the respondents for this study:

The Household Heads. The primary respondent in the Barangay-Connect. This individual is responsible for making decisions for the household and will provide detailed information on different demographic data.

Barangay Officials. Barangay officials or community leaders will be included as respondents to provide additional context and information that cannot be collected from individual households.

Local Government Unit (LGU) Representatives. Representatives from the Local Government Unit (LGU) will also be respondents in certain aspect of study.

External Experts and Researches. Depending on the scope of the study, external experts or researchers will also be consulted as respondents, particularly if the study requires expert knowledge in areas.

Data Gathering Procedures

A letter of request to conduct the interview was prepared. The researchers constructed 10 questionnaires, validated by the professor of the subject and chairman of the College of Information in Communications Technology (CICT) then the 10 questionnaires are distributed. The researchers conduct the research in Poblacion San Isidro, Nueva Ecija through an interview. The researchers explain to the respondents the importance of their response to the study. The researchers clarify and translate the questionnaires to the respondents so that the respondents can answer the interview with full knowledge for their responsibility as the subject of the study. During the interview, the researchers recorded and documented the answers of the respondents then the researchers rewrite and collect the information needed for developing the system. After the interview the researcher seeks an IT expert for help in determining the appropriate programming tools to be used and developing the system. Based on the data, researchers come up with conclusions and recommendations for this study.

Research Instruments

In the study entitled Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija, the researchers utilized three distinct types of structured questionnaires as the primary research instruments. The following are the questionnaires that were designed for the study:

1. *Questionnaire for Barangay Officials and Staff.* This instrument focused on gathering feedback from those who will directly operate the system. Questions were aligned with ISO 25010 characteristics such as functional suitability (accuracy and completeness of features for profiling), usability (ease of use, user interface design), reliability (consistency of system operation), and security (protection of household data). The responses provided insights into whether the system adequately supports day-to-day barangay operations and data management processes.
2. *Questionnaire for Heads of Households.* This instrument was intended for community members whose profiles will be recorded in the system. It focused on usability (ease of accessing and providing information), compatibility (system accessibility across different devices), and security (confidence in the safety and privacy of personal data). In addition, it included questions on functional suitability to ensure that the profiling system captures the actual needs of households. The inclusion of open-ended questions allowed residents to recommend improvements and raise concerns about data protection and service accessibility.
3. *Questionnaire for IT Experts.* This instrument was designed to assess the system from a technical and professional perspective. It covered all eight ISO 25010 quality characteristics, with emphasis on performance efficiency (response time,

resource utilization), maintainability (ease of system updates and modifications), and portability (adaptability across platforms and environments). IT experts were also asked to provide evaluations on security and reliability, ensuring that the system follows best practices in information security and system sustainability. Their expert feedback validated whether the system aligns with recognized software quality standards.

Data Analysis Technique

To evaluate the effectiveness and usability of the Barangay-Connect Household Profiling System, the study employed a combination of quantitative and qualitative data analysis techniques. These methods are designed to correspond directly to the Statement of the Problem (SOP) and provide a systematic interpretation of the data collected throughout the development and assessment phases of the project.

For Statement of the Problem 1, which focuses on the design and features of the Barangay-Connect system, the analysis was about the application of the Software Development Life Cycle (SDLC) process. Each development phase, including requirement gathering, design, implementation, and testing, described in relation to how it addressed the identified needs. Screenshots, system flow diagrams, and feature breakdowns were presented to support this.

For Statement of the Problem 2, which evaluates the technical quality of the system, the study applied descriptive statistics to survey results. This includes computing for frequency, percentage, and weighted mean to analyze user feedback regarding system

interface, functionality, accessibility, and reliability. The analysis was organized into tables and graphs to clearly illustrate users' satisfaction levels.

For Statement of the Problem 3 and 4, which assesses the system's impact on service delivery – the study will compare the responses from barangay officials and heads of household before and after using the system. Feedback from key users will be interpreted to validate whether the system addresses the original inefficiencies.

Each data analysis strategy was carefully aligned with the study's objectives and research problems. By integrating both statistical interpretation and system development documentation, the study ensures a comprehensive and evidence-based presentation of findings. This dual approach not only measures technical success but also reflects the practical value and user experience of the Barangay-Connect system.

Responses and Scoring Mode

Respondents will evaluate the Barangay-Connect: Barangay Household Profiling System++ using different scale for each parameter.

For the **Functional Suitability**, the scale was:

Table 1. Functional Suitability Scale

Scale	Qualitative Description	Descriptors
5	Very Functional	The system provides all required functions, delivers accurate results, and fully supports barangay profiling needs.
4	Functional	The system provides most functions and results are generally accurate, with only minor gaps.
3	Needs Improvement	The system provides some functions but misses important features; results are adequate but not always precise.

2	Poor	The system lacks many essential functions, with frequent inaccuracies.
1	Very Poor	The system does not provide the necessary functions for profiling and fails to deliver accurate results.

For **Performance Efficiency**, the following scale were used:

Table 2. Performance Efficiency Scale

Scale	Qualitative Description	Descriptors
5	Very Efficient	The system operates very fast, responds instantly, and uses resources efficiently even with many users.
4	Efficient	The system operates well and responds quickly, though minor delays may occur.
3	Needs Improvement	The system performs adequately but sometimes slows down or consumes more resources than expected.
2	Poor	The system often lags, delays responses, and consumes excessive resources.
1	Very Poor	The system is consistently slow, unresponsive, and resource-heavy.

For **Compatibility**, the following scale were used:

Table 3. Compatibility Scale

Scale	Qualitative Description	Descriptors
5	Very Compatible	The system works seamlessly across devices, browsers, and platforms without conflicts.
4	Compatible	The system operates well and responds quickly, though minor delays may occur.
3	Needs Improvement	The system performs adequately but sometimes slows down or consumes more resources than expected.
2	Poor	The system often lags, delays responses, and consumes excessive resources.
1	Very Poor	The system is consistently slow, unresponsive, and resource-heavy.

For **Usability**, the following scale were used:

Table 4. Usability Scale

Scale	Qualitative Description	Descriptors
5	Very Usable	The system is very user-friendly, easy to learn, and has a clear, intuitive interface.
4	Usable	The system is generally easy to use, requiring minimal training.
3	Needs Improvement	The system is moderately usable but requires effort or practice to learn.
2	Poor	The system is difficult to use, confusing, and requires significant guidance.
1	Very Poor	The system is very hard to use, with a poor interface and unclear instructions.

For **Reliability**, the following scale were used:

Table 5. Reliability Scale

Scale	Qualitative Description	Descriptors
5	Very Reliable	The system consistently operates without errors, downtime, or data loss.
4	Reliable	The system is generally reliable, with only occasional issues.
3	Needs Improvement	The system works most of the time but sometimes fails or loses data.
2	Poor	The system frequently encounters errors and is not dependable.
1	Very Poor	The system is highly unreliable, often unavailable, and prone to data loss.

For **Security**, the following scale were used:

Table 6. Security Scale

Scale	Qualitative Description	Descriptors
5	Very Secure	The system fully protects data, prevents unauthorized access, and ensures user privacy.
4	Secure	The system is secure, with only minor risks.
3	Needs Improvement	The system has basic security features but may be vulnerable.
2	Poor	The system has weak protection and is at risk of unauthorized access.
1	Very Poor	The system is not secure and exposes sensitive data.

For **Maintainability**, the following scale were used:

Table 7. Maintainability Scale

Scale	Qualitative Description	Descriptors
5	Very Maintainable	The system is easy to update, debug, and enhance without affecting other functions.
4	Maintainable	The system can be maintained and updated with minor effort.
3	Needs Improvement	The system can be updated but with some difficulty or risk.
2	Poor	The system is hard to maintain, with frequent complications during updates.
1	Very Poor	The system is extremely difficult to update or improve.

For **Portability**, the following scale were used:

Table 8. Potability Scale

Scale	Qualitative Description	Descriptors
5	Very Portable	The system can be easily installed, transferred, and adapted to other environments or barangays.

4	Portable	The system can be transferred and adapted with little adjustment.
3	Needs Improvement	The system can be transferred but requires considerable effort.
2	Poor	The system is difficult to transfer and adapt.
1	Very Poor	The system cannot be transferred or adapted to other environments.

Statistical Analysis of Data

In the context of software engineering, software quality measures how well software is specified and, how well the software conforms to the specifications. Respondents will score the developed system according to its quality. As part of this assessment, a rubric scoring was used to analyze and interpret respondents' mean scores. Wherein, a high mean rating categorically states a very positive view and a low mean rating implies a poor outcome.

The following rubrics were defined for the evaluation of the developed system:

Table 9: The Rubrics for the Evaluation of the Developed System

Numerical Rating	Qualitative Rating	Description
4.20 – 5.00	Excellent	The developed system meets all the quality standards of software development. No or very minimal modification is required.
3.40 – 4.19	Very Good	The developed system meets almost all the quality standards of software development. Minimal modification is required.
2.60 -3.39	Good	The developed system meets some of the quality standards of software development. Some revisions are required.
1.80 – 2.59	Fair	The developed system fails to meet the quality standards of software development. Major revisions are required.

1.00 – 1.79	Poor	The developed system fails to meet the quality standards of software development. Needs to be redone to serve its purpose.
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Chapter III

Results and Discussions

This chapter serves as the core where the outcomes of the system's development and evaluation are presented, analyzed, and interpreted. For the study entitled Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija, this section plays a vital role in demonstrating how the proposed system successfully addressed the identified problems in managing and organizing household records within the barangay. It provides the evidence that the system not only functions as intended but also contributes to improving the efficiency, accuracy, and accessibility of barangay data management.

1. Development of BARANGAY-CONNECT: Barangay Household Profiling System++ for San Isidro, Nueva Ecija

The development of the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija underwent the stages of the System Development Cycle:

1.1. Planning Phase

The planning phase of the study focused on laying the groundwork for the development of the Barangay-Connect: Barangay Household Profiling System++. This stage involved identifying the needs of the barangay, clarifying the objectives of the project, and evaluating the feasibility of its implementation. The study was initiated after recognizing the inefficiencies of the existing manual process of household profiling in San Isidro, Nueva Ecija. Barangay officials currently rely on paper-based records, which often result in delays, data inaccuracies, and difficulties

in retrieving household information. These challenges underscored the necessity of creating a centralized and automated profiling system that could streamline operations and provide reliable data for decision-making.

The planning phase also included the identification of necessary resources such as human resources, technical tools, and financial allocations. Risks were carefully considered, including possible resistance from staff who may not be accustomed to digital systems, issues with internet connectivity, and concerns regarding data privacy and security. Addressing these risks early on guided the formulation of strategies to ensure smooth implementation, such as providing training sessions, incorporating offline access features, and applying strict data security measures.

By the end of the planning phase, the researchers had clearly defined the project's objectives, scope, feasibility, and resource requirements. This phase produced essential deliverables such as the project proposal, feasibility study, and initial scheduling, which together served as the foundation for moving forward to the analysis and design phases. Through careful planning, the study ensured that the Barangay-Connect system would be both technically feasible and socially acceptable, ultimately contributing to more efficient and reliable household data management in San Isidro, Nueva Ecija.

Gantt Chart

Table 10. Gantt Chart of the Activities for the Development of the Barangay-Connect

Activities	Aug				Sep				Oct				Nov				Dec			
Planning																				
Designing																				
Development																				
Testing																				
Deployment																				
Review																				
Maintenance																				

The Gantt Chart for Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija illustrates the systematic timeline and distribution of activities undertaken throughout the development of the system. The project was structured following the Software Development Life Cycle (SDLC), which ensured an organized, efficient, and iterative process. The entire project was completed within five months (20 weeks), beginning from the planning phase and concluding with the documentation and final defense.

1.2. Designing Phase

The designing phase of the Barangay-Connect: Barangay Household Profiling System++ focused on transforming the requirements gathered during the planning stage into a structured system design that would serve as the blueprint for development. At this stage, the researchers carefully considered both the functional and non-functional requirements of the project to ensure that the system would not

only meet the needs of the barangay but also align with the standards of usability, efficiency, and security.

The design began with the establishment of the system architecture, which outlined the relationship between the client interface, the application logic, and the database management system. The system was conceptualized as a web-based platform to provide accessibility and scalability, allowing barangay officials to manage household information efficiently. The database design was created to store demographic, socioeconomic, and housing details, ensuring that data relationships were normalized to minimize redundancy and preserve accuracy. An Entity-Relationship Diagram (ERD) was constructed to illustrate the connections among entities such as households, residents, and barangay records, providing a clear representation of how data would be organized.

Furthermore, the design phase included defining the system's performance requirements. The system was designed to process and retrieve household records quickly, even under multiple concurrent users. Compatibility was also ensured so that the system could function across different devices and web browsers, maximizing accessibility for barangay personnel. In addition, scalability was considered in the design to allow the system to be adapted for use by other barangays in the future without requiring major modifications.

By the conclusion of the designing phase, a comprehensive system model had been produced, including the architectural framework, database schema, data flow diagrams, and interface mockups. This phase ensured that the *Barangay-Connect* system had a clear and detailed design that could guide the development

process. With a well-prepared blueprint in place, the project was ready to move into the development and implementation phase, where the design would be translated into a working software application that could enhance household profiling and data management in San Isidro, Nueva Ecija.

Use Case

The Barangay-Connect: Barangay Household Profiling System++ is a web-based information system designed to automate the management of household records within Barangay San Isidro, Nueva Ecija. It enables barangay officials to efficiently record, update, search, and generate reports on household data. The system aims to improve accuracy, transparency, and accessibility of information for effective local governance.

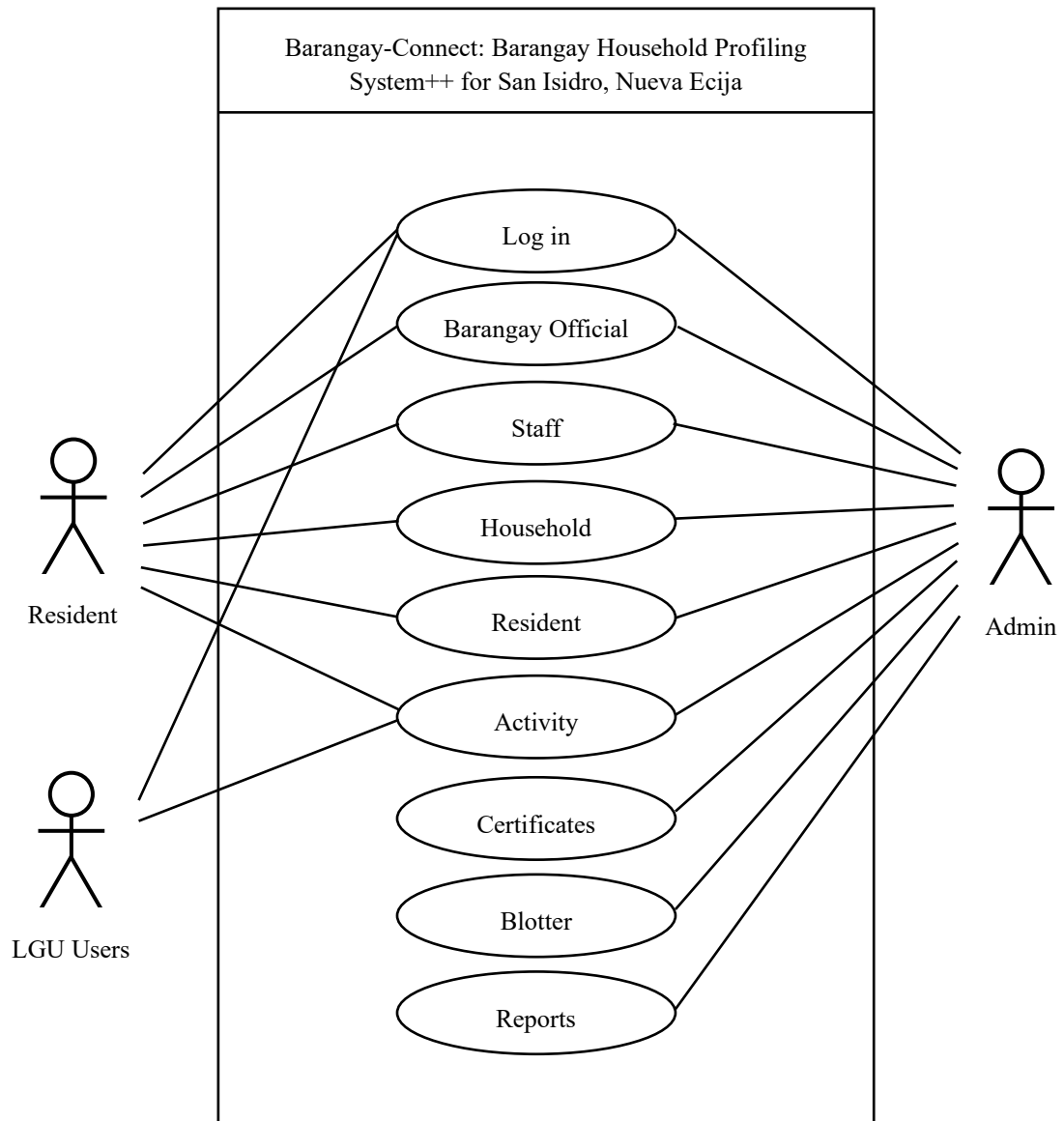


Figure 4. Use Case Diagram of Barangay-Connect

Context Diagram Level 0

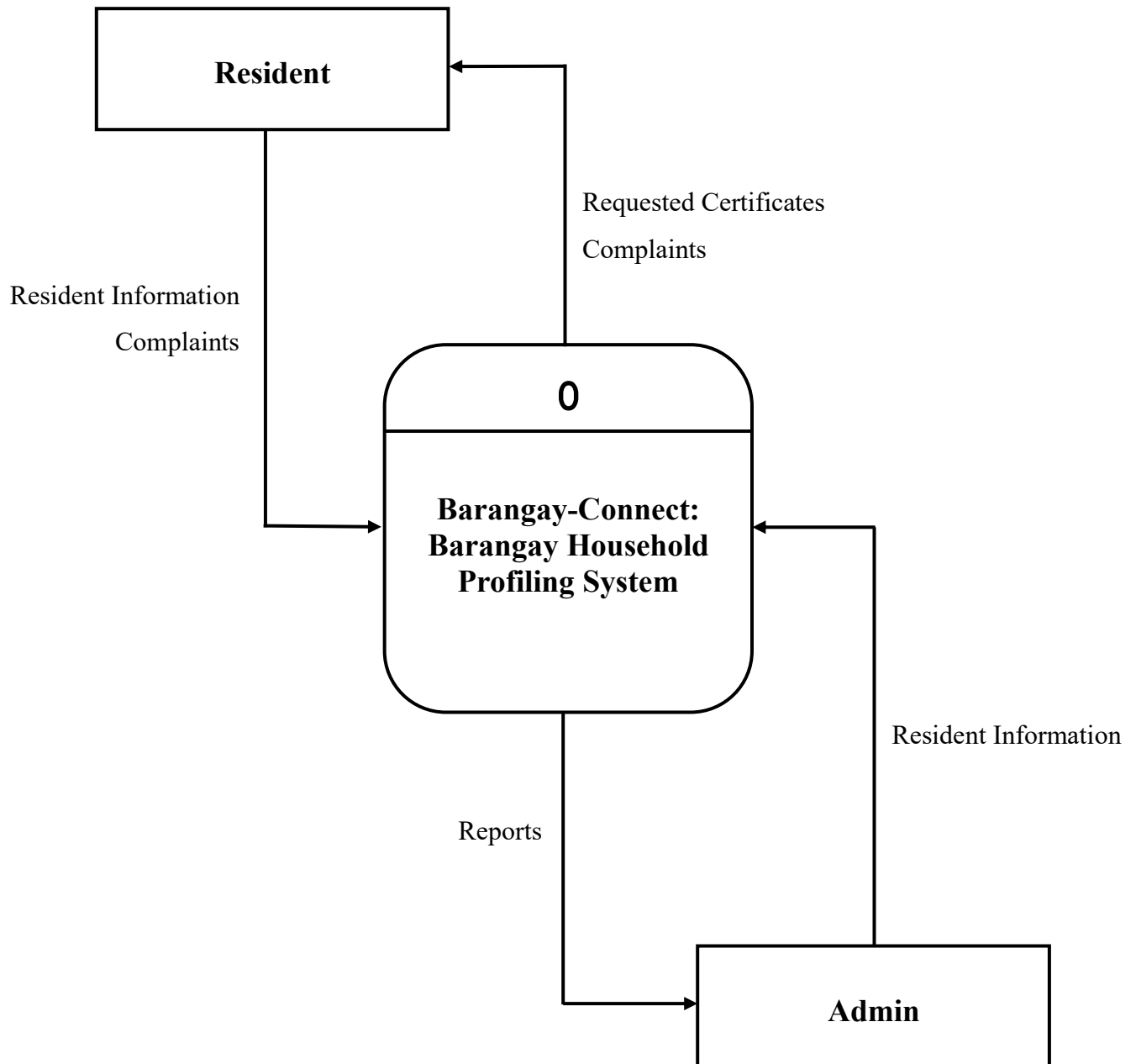


Figure 5. Context Diagram Level 0

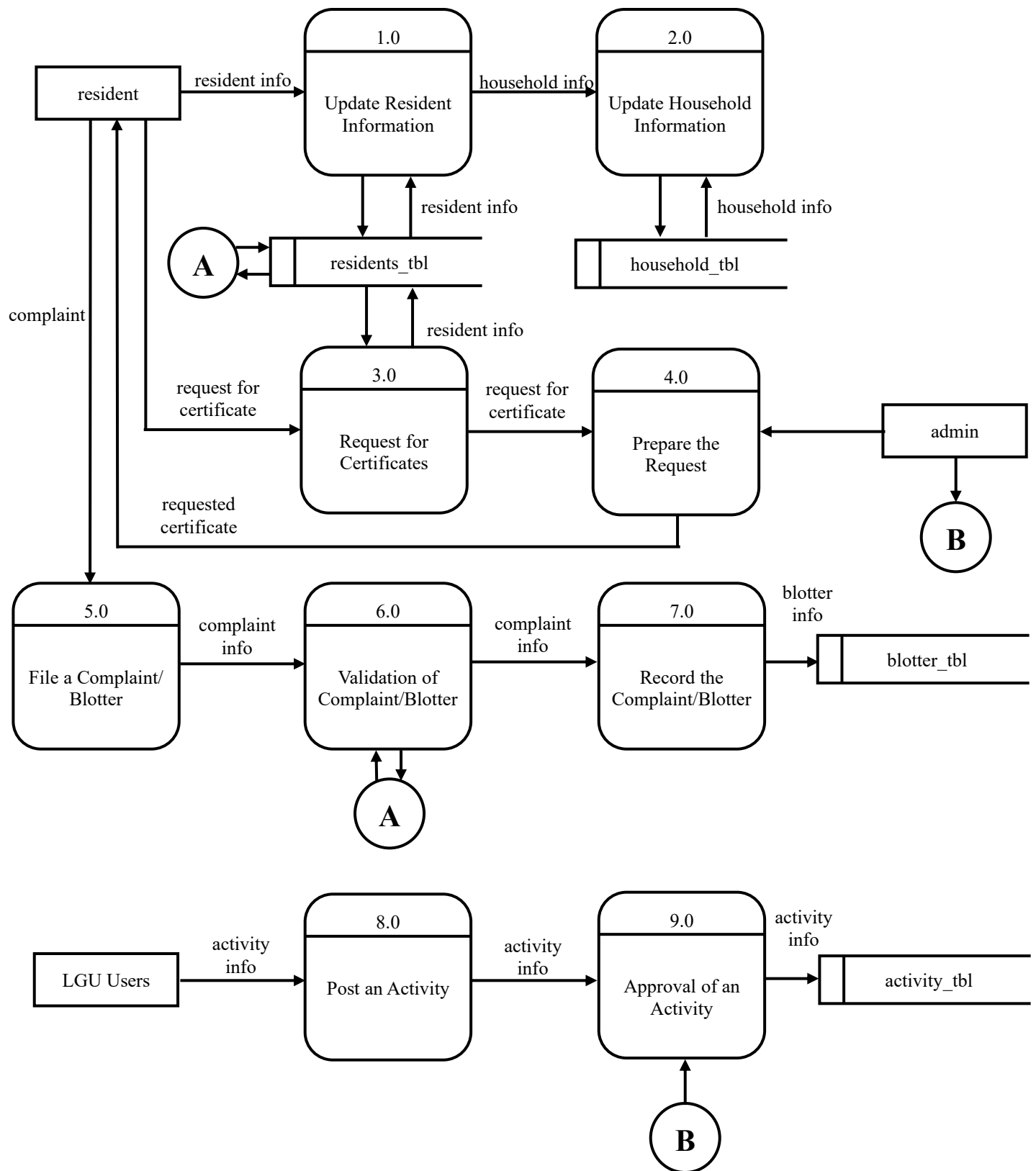


Figure 6. Context Diagram Level 1

1.3. Development Phase

The development phase of the Barangay-Connect: Barangay Household Profiling System++ marked the actual construction of the system, where the designs and specifications prepared in the earlier phases were translated into functional software. This stage involved programming, database creation, and the integration of system components to produce a working application that could be tested and evaluated.

The system was developed using web-based technologies to ensure accessibility, scalability, and user convenience. The researchers employed a modular development approach, which meant that individual features of the system—such as household registration, resident profiling, report generation, and account management—were coded and tested separately before being integrated into the main system. This approach minimized errors and allowed for the early detection of problems within each module.

Database implementation was a critical part of the development process. The Entity-Relationship Diagram (ERD) and database schema designed in the earlier stage were converted into a structured relational database that stored household demographic, socioeconomic, and housing information. Proper normalization techniques were applied to eliminate redundancy and ensure data integrity. Queries and stored procedures were also developed to enable fast retrieval and updating of household records, which was essential for efficient barangay operations.

During the coding process, the researchers placed strong emphasis on user interface development. The system was designed with a simple and intuitive interface to accommodate barangay officials and staff who may not be highly proficient with technology. Forms, input fields, and buttons were carefully arranged and labeled to guide users in performing their tasks, such as entering household details, searching for records, or generating reports. The design also incorporated validation features to minimize input errors and ensure accuracy of data entry.

The development phase was carried out in iterative cycles, allowing continuous improvement and refinement of the system. After each module was developed, it was tested against the requirements, and adjustments were made when inconsistencies or errors were identified. This iterative process ensured that the final version of the system met the functional requirements established during the planning and analysis phases.

By the end of the development phase, the Barangay-Connect system was fully constructed and ready for deployment in the barangay setting. The system could handle household profiling tasks efficiently, generating accurate reports, and providing barangay officials with a reliable tool for data management. This phase provided the concrete output of the research, transforming the conceptual design into an operational system that addressed the challenges of manual profiling in San Isidro, Nueva Ecija.

Application Programs

The Application Programs of the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija were developed using PHP

(Hypertext Preprocessor) as the primary programming language and MySQL as the relational database management system (RDBMS). This combination was chosen for its proven reliability, scalability, and compatibility with web-based applications that require efficient handling of large datasets, such as household and resident records. The integration of PHP and MySQL provides a dynamic, interactive, and secure environment for managing barangay data and delivering responsive functionalities to users.

The system's architecture follows a client-server model, wherein users interact with the front-end interface developed in PHP, HTML, CSS, and JavaScript, while the back-end logic communicates with the MySQL database to process, store, and retrieve data. Each application program or module was carefully designed to address a specific function in the barangay's household profiling operations.

Hypertext Preprocessor (PHP)

PHP (Hypertext Preprocessor) played a crucial role in the development of Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija as the primary server-side scripting language used to build the system's back-end functionality. It served as the core technology that enabled dynamic data processing, database communication, and real-time interaction between users and the system. The use of PHP ensured that the application was efficient, secure, and scalable, aligning with the project's goal of providing a reliable and responsive household profiling system.

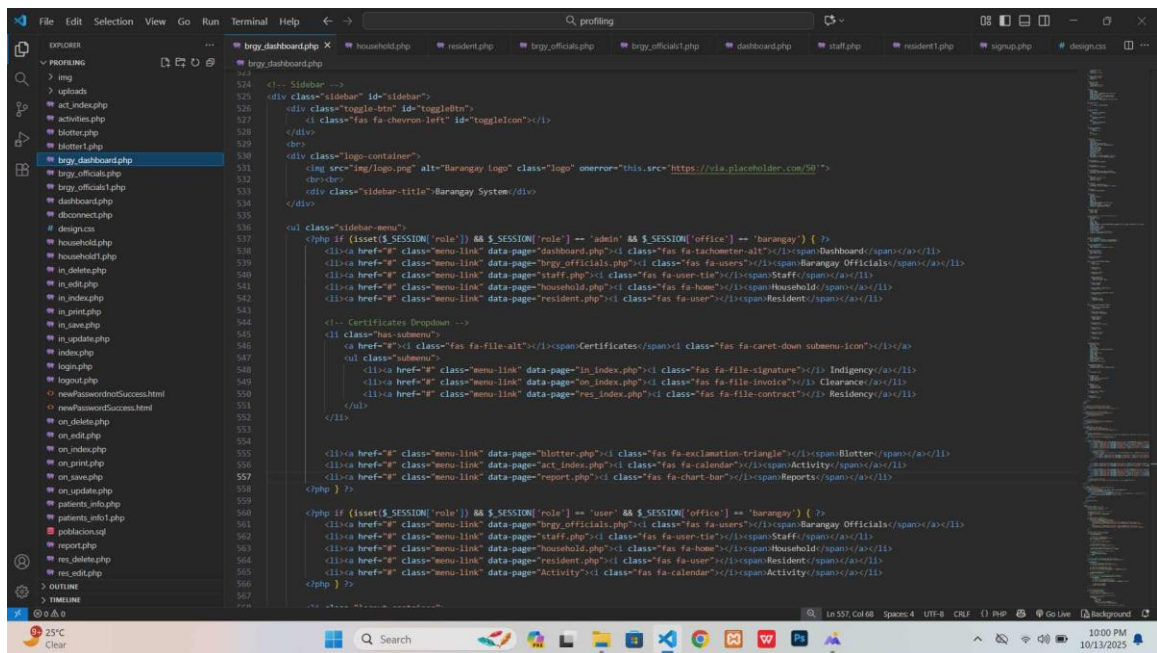


Figure 7. Visual Studio Code Environment with PHP File

MySQL (XAMP)

MySQL, accessed through the WAMP (Windows, Apache, MySQL, PHP) server environment, played a vital role as the database management system (DBMS) in developing Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija. It served as the backbone of the system's data storage, organization, and retrieval processes. By utilizing MySQL's robust relational database capabilities within the WAMP server, the developers were able to design an efficient, secure, and scalable data management solution that ensured accurate and organized household profiling for the barangay.

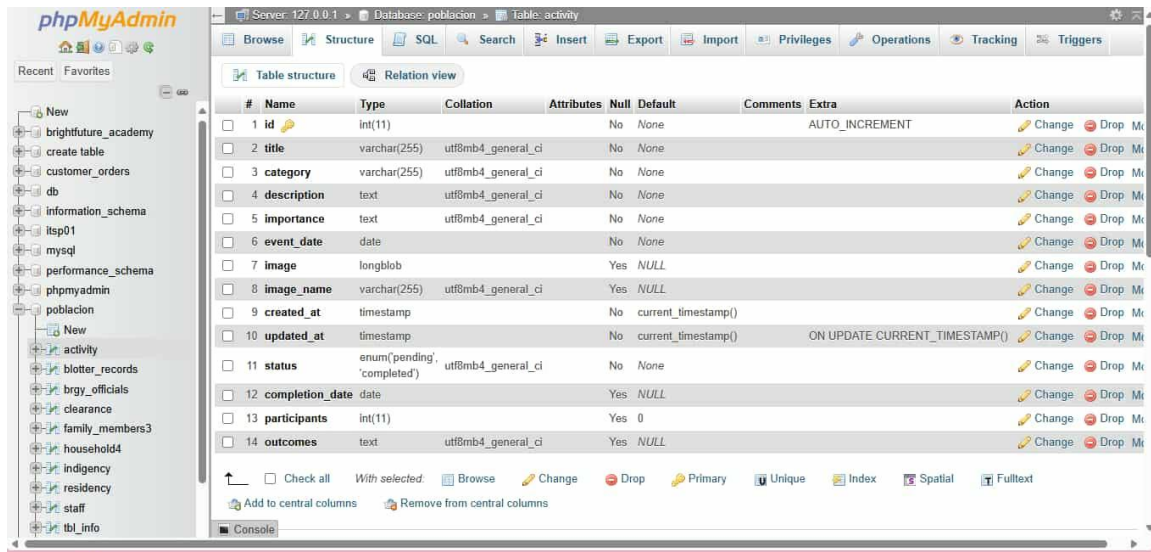


Figure 8. phpMyAdmin Environment

Database Normalization

Database normalization is a critical process in the design of the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija, as it ensures the accuracy, consistency, and efficiency of the data stored in the system. Normalization is the systematic method of organizing data in a database to minimize redundancy and dependency by dividing large tables into smaller, related ones. This process promotes data integrity and enhances the performance of the system in handling queries, updates, and reports.

The normalization of the Barangay-Connect database followed the three fundamental normal forms (1NF, 2NF, and 3NF) to ensure optimal structure and logical data organization.

Through the normalization process, the Barangay-Connect: Barangay Household Profiling System++ achieved a well-structured and relational database design capable of managing large volumes of data efficiently. It allows faster retrieval of records, accurate reporting of household demographics, and easy maintenance of the database structure. Moreover, the normalized design ensures that the system can be seamlessly integrated with future modules, such as disaster response management, social assistance tracking, and barangay statistical reporting.

Unnormalized Data

```
lastname  
firstname  
middlename  
ext  
place  
birth  
age  
sex  
civil_status  
citizen  
occupation  
work  
household_size  
family_lastname  
family_firstname  
family_middlename  
family_ext  
family_relationship  
family_place  
family_birth  
family_age  
family_sex  
family_civil_status  
family_citizen  
family_occupation  
family_work
```

First Normal Form (1NF)

household
lastname
firstname
middlename
ext
place
birth
age
sex
civil_status
citizen
occupation
work
household_size

family_members
family_lastname
family_firstname
family_middlename
family_ext
family_relationship
family_place
family_birth
family_age
family_sex
family_civil_status
family_citizen
family_occupation
family_work

Second Normal Form (2NF)

household
lastname firstname middlename ext place birth age sex civil_status citizen occupation work household_size

family_members
family_lastname family_firstname family_middlename family_ext family_relationship family_place family_birth family_age family_sex family_civil_status family_citizen family_occupation family_work

blotter_records
complaint_name respondent_name incident_type incident_location incident_date incident_time complaint_details barangay_officer

staff
position first_name last_name suffix gender date_of_birth age place_of_birth civil_status religion citizenship contact_number hire_date

barangay_officials
complaint_name respondent_name incident_type incident_location incident_date incident_time complaint_details barangay_officer

Third Normal Form (3NF)

household
household_id(PK) lastname firstname middlename ext place birth age sex civil_status citizen occupation work household_size

family_members
family_id(PK) household_id(FK) family_lastname family_firstname family_middlename family_ext family_relationship family_place family_birth family_age family_sex family_civil_status family_citizen family_occupation family_work

blotter_records
blotter_id(PK) complaint_name respondent_name incident_type incident_location incident_date incident_time complaint_details staff_id(FK)

staff
staff_id(PK) position first_name last_name suffix gender date_of_birth age place_of_birth civil_status religion citizenship contact_number hire_date

barangay_officials
official_id(PK) complaint_name respondent_name incident_type incident_location incident_date incident_time complaint_details barangay_officer

Entity-Relationship Diagram

The Entity-Relationship Diagram (ERD) serves as the conceptual blueprint of the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija. It illustrates the logical structure of the database and defines how data elements are interrelated within the system. The ERD was developed to ensure that all data entities, their attributes, and relationships accurately reflect the operational needs of the barangay in managing household and resident information.

The relationships between entities in the ERD are established using appropriate cardinalities—such as one-to-one, one-to-many, and many-to-many—depending on the logical associations among data sets. Each relationship is further strengthened using primary and foreign keys, ensuring referential integrity, and preventing data inconsistencies across the database.

Overall, the Entity-Relationship Diagram provides a clear and organized visualization of the Barangay-Connect: Barangay Household Profiling System++ database architecture. It ensures that every data interaction is systematically mapped, supporting efficient data retrieval, accurate reporting, and secure information management. By grounding the system's design on this ERD, the project guarantees a structured, reliable, and scalable data model that aligns with the operational needs of the barangay government of San Isidro, Nueva Ecija.

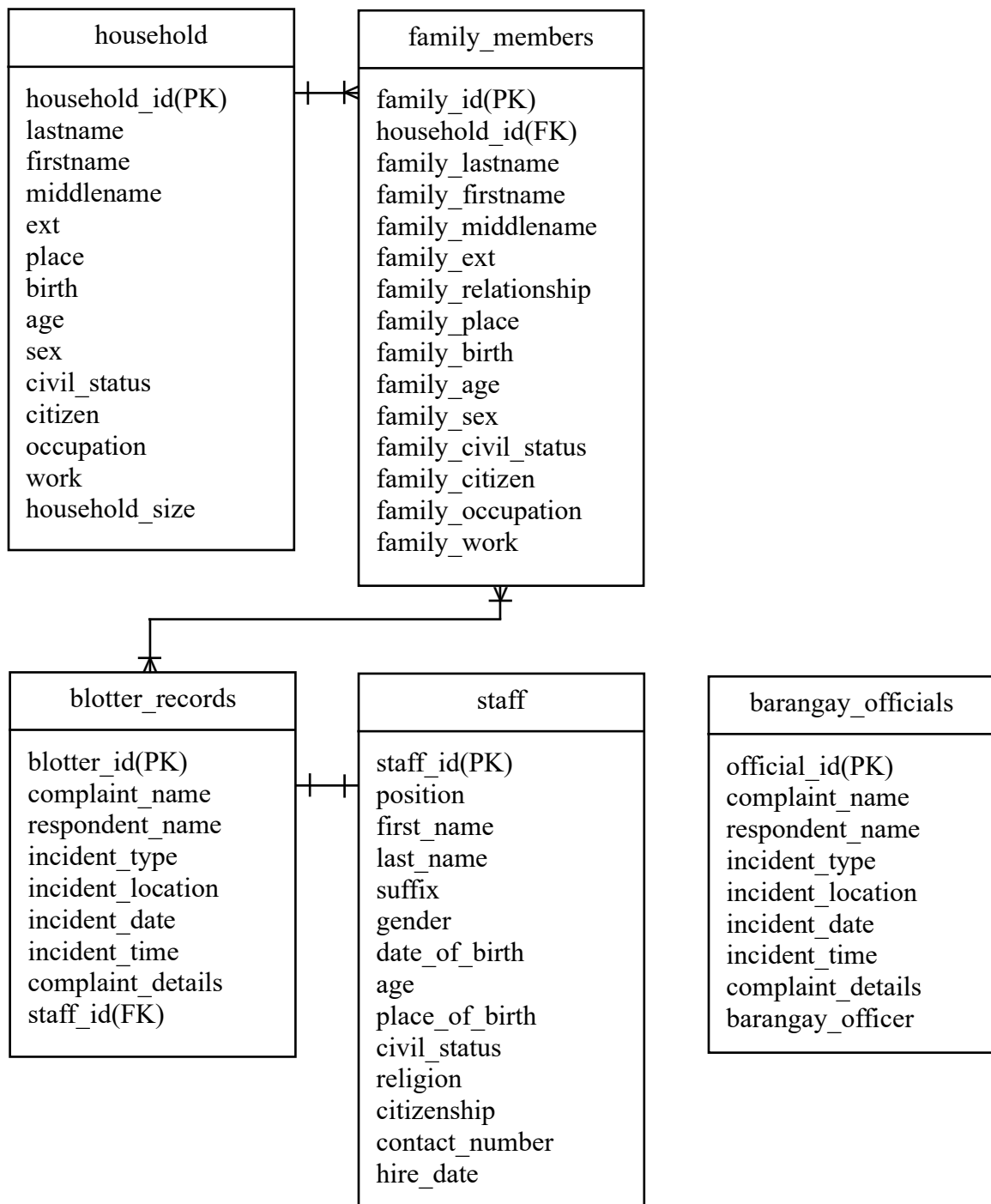


Figure 9. The Entity-Relationship Diagram of Barangay-Connect

Data Dictionary

The Data Dictionary of the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija serves as the central reference for all data elements used throughout the system. It provides a detailed description of the structure, characteristics, and purpose of each data field within the database. By defining the names, types, formats, and relationships of data elements, the data dictionary ensures consistency, accuracy, and clarity in data management across all modules of the system.

The data dictionary was developed alongside the Entity-Relationship Diagram (ERD) and database normalization process to ensure alignment between the system's logical and physical database design. Each entry in the data dictionary specifies the field name, data type, field size or length, description, and associated table. This structure facilitates efficient database maintenance, minimizes redundancy, and provides developers and users with a clear understanding of how data is organized within the system.

activity table is used to store information about barangay activities and events		
Field Name	Data Type	Description
id (primary)	int(11)	Unique identifier for activity.
title	varchar(255)	The title of the activity.
category	varchar(255)	The category the activity belongs to
description	text	A detailed description of the activity.
importance	text	Describes the importance of event.
event_date	date	The scheduled date for the event.
image	longblob	A documentation for the event.
image_name	varchar(255)	The name of image.

created_at	timestamp	The date of when created.
updated_at	timestamp	The date of when updated.
status	enum('pending','completed')	The current status of the activity.
completion_date	date	The date of completion.
participants	int(11)	The number of participants.
outcomes	text	The outcome of the activity.

blotter_records table is used for storing information of blotter cases filed in the barangay.		
Field Name	Data Type	Description
id (primary)	int(11)	Unique identifier for blotter activity.
case_number	varchar(50)	The assigned case number for the incident.
complainant_name	varchar(100)	The name of the person filing the complaint.
respondent_name	varchar(100)	The name of the person the complaint is against
incident_type	varchar(50)	The type of incident being reported.
incident_location	varchar(200)	Where the incident took place.
incident_date	text	The date the incident occurred.
incident_time	time	The time the incident occurred.
complaint_details	text	A detailed account of the complaint.
barangay_officer	varchar(100)	A barangay officials or staff.
status	enum('pending','Resolved')	The current status of the case.

brgy_officials table contains information about the barangay officials.		
Field Name	Data Type	Description
id (primary)	int(11)	Unique identifier for brgy_officials.
image_path	varchar(200)	A path where the image will be uploaded.
position	varchar(100)	The official's position or title.
first_name	varchar(100)	The first name of the official.
last_name	varchar(100)	The last name of the official.
middle_name	varchar(100)	The middle name of the official.
suffix	varchar(50)	The name extension (e.g., Jr., Sr., etc.)
gender	enum('Male','Female')	The gender of the official.
date_of_birth	text	The date of birth of the official.
age	int(100)	The age of official.

place_of_birth	varchar(50)	The place of birth of the official.
civil_status	enum('Single','Married','Widowed','Separated','Divorced')	The civil status of the official.
religion	varchar(100)	The religion of the official.
citizenship	varchar(100)	The citizenship of the official.
contact_number	varchar(20)	The contact number of the official.

clearance table is used to store information for issuing barangay clearances.		
Field Name	Data Type	Description
id (primary)	int(11)	Unique identifier for clearance.
resident_name	varchar(100)	The name of the resident requesting the clearance
address	text	The address of the resident.
dob	date	Date of birth of the resident.
pob	varchar(100)	Place of birth of the resident.
height	varchar(20)	The height of the resident.
weight	varchar(20)	The weight of the resident.
civil_status	enum('Single', 'Married', 'Widowed', 'Separated', 'Divorced')	The civil status of the resident
finding	text	The findings of the resident
purpose	varchar(255)	The reason for requesting the clearance.
date_issued	date	The date the clearance was issued

family_members3 table stores Information about the members of the household.		
Field Name	Data Type	Description
id (primary)	int(11)	Unique identifier for family_member.
household_id	int(11)	Identifier linking to the household.
lname	varchar(50)	The last name of the family member.
fname	varchar(50)	The first name of the family member.
mname	varchar(50)	The middle name of the family member.
ext	varchar(100)	The name extension (e.g., Jr.)
relationship	varchar(100)	The member's relationship to the head of the household.
place	varchar(100)	Place of birth of the family member.
birth	text	Date of birth of the family member.
age	int(11)	Age of the family member.

sex	enum('Male', 'Female')	The gender of the family member.
civil_status	enum('Single', 'Married', 'Widowed', 'Separated')	The civil status of the family member.
citizen	varchar(50)	The citizenship of the family member.
occupation	varchar(50)	The occupation of the family member.
work	enum('Employed', 'Unemployed', 'PWD', 'OFW', 'Solo Parent')	Work of the family member.

household4 table contains information about each household, typically focusing on the head of the household.		
Field Name	Data Type	Description
id (primary)	int(11)	Unique identifier for household.
user_id	int(11)	Identifier for user.
Iname	varchar(50)	The last name of the head of the household.
fname	varchar(50)	The first name of the head of the household.
mname	varchar(50)	The middlename of the head of the household.
ext	varchar(100)	The name extension (e.g., Jr.)
place	varchar(100)	The place of birth of the household.
birth	text	Date of birth of the household.
age	int(11)	The age of the head of the household.
sex	enum('Male', 'Female')	The gender of the head of the household.
civil_status	enum('Single', 'Married', 'Widowed', 'Separated', 'Divorced')	The civil status of the head of the household.
citizen	varchar(50)	Citizenship of the head of the household.
occupation	varchar(100)	The occupation of the head of the household.
work	enum('Employed', 'Unemployed', 'PWD', 'OFW', 'Solo Parent')	Work of the head of the household.
household_size	int(11)	The total numbers of members in the household.
status	enum('Active', 'Inactive')	The status of the household record

indigency table stores records for certificates of indigency.		
Field Name	Data Type	Description
id (primary)	int(11)	Unique identifier for indigency.
resident_name	varchar(100)	The name of the resident.
date_issued	datetime	The date and time the certificate was issued

residency table is used for storing information to issue certificates of residency.		
Field Name	Data Type	Description
id (primary)	int(11)	Unique identifier for residency.
resident_name	varchar(255)	The name of the resident.
address	varchar(255)	The address of the resident.
date_of_birth	date	Date of birth of the resident.
place_of_birth	varchar(255)	Place of birth of the resident.
civil_status	varchar(50)	Civil status of the resident.
gender	varchar(20)	Gender of the resident.
date_issued	date	The date the certificate was issued

staff table contains information about the barangay staff members.		
Field Name	Data Type	Description
id (primary)	int(11)	Unique identifier for staff.
position	varchar(100)	The staff's position or title.
first_name	varchar(100)	The first name of the staff member.
middle_name	varchar(100)	The middle name of staff member.
last_name	varchar(100)	The last name of the staff member.
suffix	varchar(50)	The name extension (e.g., Jr.).
gender	enum('Male', 'Female')	The gender of the staff member.
date_of_birth	text	The date of birth of the staff member.
age	int(100)	The age of staff member.
place_of_birth	varchar(50)	The place of birth of the staff member.
civil_status	enum('Single', 'Married', 'Widowed', 'Separated')	The civil status of the staff member.
religion	varchar(100)	The religion of the staff member.
citizenship	varchar(100)	The citizenship of the staff member.
contact_number	varchar(20)	The contact number of the staff member.
hire_date	varchar(100)	The date the staff member was hired
status	Enum ('Active', 'Inactive')	The status of staff member.

tbl_info table is used for storing general personal information about the residents.		
Field Name	Data Type	Description
id (primary)	int(11)	Unique identifier for tbl_info.
lname	varchar(100)	The last name of the person.
fname	varchar(100)	The first name of the person.
mname	varchar(100)	The middle name of the person.
ext	varchar(50)	The name extension (e.g., Jr.)
place_of_birth	varchar(100)	The person's place of birth.
date of birth	text	The person's date of birth.
age	int(11)	The age of the person.
gender	enum('Male', 'Female')	The gender of the person.
civil status	enum('Single', 'Married', 'Widowed', 'Separated')	The civil status of the person.
citizenship	varchar(50)	The citizenship of the person.
occupation	varchar(50)	The person's occupation
contact_number	Int(11)	The contact number of the person.

tbl_users table is used for storing user account information for system access.		
Field Name	Data Type	Description
id (primary)	int(11)	Unique identifier for tbl_users.
account type	enum('Primary', 'Secondary')	The type of user account.
username	varchar(255)	The username for logging into the system
password	varchar(255)	The user's password
role	enum('admin', 'user')	The role assigned to the user
office	enum('barangay', 'rhu')	The office the user is in.

1.4. Testing Phase

The testing phase of the Barangay-Connect: Barangay Household Profiling System++ was conducted to ensure that the system performed according to its intended functions and met the requirements defined during the earlier phases of development. This stage served as the quality assurance process, where the system was subjected to different levels of testing to identify defects, verify functionality,

and validate overall performance. The purpose of testing was not only to detect errors but also to confirm that the system was reliable, secure, and user-friendly before deployment in the barangay setting.

The testing process began with unit testing, where individual modules such as household registration, resident profiling, report generation, and account management were evaluated separately. Each function was checked for accuracy, proper execution, and compliance with design specifications. For example, the registration module was tested to determine whether it could accurately save, update, and retrieve household records without errors. This stage helped ensure that every component of the system was working as expected before integration.

Once the modules passed unit testing, integration testing was conducted to verify that the different components of the system functioned smoothly when combined. This stage focused on the flow of data between the database and the application interface, as well as the interaction between various modules. Scenarios such as updating household information and immediately generating a barangay report were tested to confirm that the system-maintained data consistency and produced accurate outputs.

After integration testing, system testing was carried out to evaluate the overall behavior of the system under real-world conditions. The researchers tested the system against the predefined requirements, such as response time, accuracy of reports, and accessibility across different devices and browsers. Special attention was given to security features, where login authentication and role-based access were tested to ensure that only authorized personnel could access sensitive

household information. Data backup and recovery functions were also simulated to verify the system's ability to preserve information during unexpected failures.

Finally, user acceptance testing (UAT) was performed by engaging actual barangay officials, staff, and selected heads of households. These users were asked to interact with the system by performing typical tasks, such as adding records, searching for household data, and generating reports. Their feedback provided valuable insights into the system's usability, clarity of navigation, and overall effectiveness in addressing the barangay's profiling needs. Minor issues identified during this stage were corrected to ensure that the system was ready for deployment.

Through these layers of testing, the researchers confirmed that the Barangay-Connect system was functional, efficient, and secure. It successfully met the requirements set out in the planning phase and demonstrated reliability in managing household data. The testing phase ensured that the final product was not only technically sound but also acceptable to its intended users, making it a dependable tool for household profiling and data management in San Isidro, Nueva Ecija.

1.5. Deployment Phase

The deployment phase of the Barangay-Connect: Barangay Household Profiling System++ marked the transition of the system from development and testing into actual use within the barangay of San Isidro, Nueva Ecija. This stage involved installing the system, configuring the environment, and ensuring that all

necessary resources were in place to support smooth operations. The primary goal of deployment was to make the system fully functional and accessible to barangay officials and staff, while minimizing disruptions to existing processes.

The process began with the installation of the system on the designated computers within the barangay office, accompanied by the proper setup of the database and user accounts. Configuration was tailored to the needs of the barangay, including the establishment of administrator and staff-level access rights based on their respective responsibilities. Security protocols such as password authentication and data backup procedures were activated during this stage to safeguard the integrity of household information.

Alongside installation, training sessions were conducted to familiarize barangay officials and staff with the system's features. The training emphasized basic operations such as registering households, updating resident information, generating reports, and navigating the interface. Since the system was designed with user-friendliness in mind, staff were able to adapt quickly, although additional guidance and manuals were provided for reference. These capacity-building activities ensured that end-users felt confident and capable of operating the system effectively.

To further guarantee a smooth transition, the deployment was carried out gradually. Pilot testing was conducted in selected households to verify the system's accuracy and performance under actual conditions before full-scale implementation. Feedback gathered during this stage allowed the researchers to refine certain aspects of the system, particularly in terms of usability and reporting

formats. Once adjustments were made, the system was officially launched for use across all households in the barangay.

Throughout deployment, technical support was made available to address any issues that arose. The researchers closely monitored the system's initial performance, ensuring that it functioned reliably and met the barangay's operational needs. Staff were encouraged to report errors or difficulties, which were promptly resolved to prevent disruptions in service delivery.

By the end of the deployment phase, the *Barangay-Connect* system had been fully integrated into the barangay's profiling operations. Household records were digitized and securely stored in the database, while officials gained a faster and more reliable means of generating reports and managing data. The successful deployment demonstrated the system's readiness for real-world application, laying the groundwork for long-term use and future enhancements.

1.6. Review Phase

The review phase of the Barangay-Connect: Barangay Household Profiling System++ served as a critical evaluation stage, where the system's overall performance, effectiveness, and impact were assessed after deployment. This phase aims to determine whether the system will meet the objectives defined during the planning stage, whether it will be addressed the challenges previously encountered in manual household profiling, and whether further refinements were necessary to enhance its use within the barangay.

During this stage, feedback will be gathered from various stakeholders, including barangay officials, staff members, and household representatives who directly interacted with the system. Their insights will be collected through interviews, surveys, and observation of system use in real-world scenarios. The evaluation will focus on core aspects of the system such as functional suitability, performance efficiency, usability, and reliability, based on the ISO 25010 software quality model. This comprehensive feedback process will allow the researchers to gain an in-depth understanding of how the system will perform in actual operations and how it will be perceived by its users.

The researchers will carefully document all feedback and observations, using them as a basis for planning future updates and maintenance activities. This process will ensure that the system remained adaptable and responsive to the evolving needs of the barangay. By reflecting on both the strengths and areas for development, the review phase not only will validate the success of the project but also highlights opportunities for continuous improvement.

In conclusion, the review phase will confirm if the Barangay-Connect system will achieve its intended purpose of modernizing household profiling in San Isidro, Nueva Ecija. It will provide faster, more accurate, and more secure management of household records while reducing the limitations of manual processes. At the same time, the review will emphasize the importance of ongoing refinement and user engagement to sustain the system's long-term relevance and effectiveness.

1.7. Maintenance and Support

The maintenance and support phase of the Barangay-Connect: Barangay Household Profiling System++ will ensure the continued functionality, reliability, and adaptability of the system after its deployment in San Isidro, Nueva Ecija. This phase will focus on sustaining the system's performance, addressing technical issues that might arise over time, and implementing improvements based on user feedback gathered during the review stage. The goal will guarantee that the system remains responsive to the evolving needs of the barangay and continue to provide accurate, secure, and efficient household profiling.

During maintenance, regular system updates will carry out to fix bugs, patch vulnerabilities, and optimize performance. These updates will ensure that the database operated smoothly, the user interface will remain responsive, and security protocols were reinforced against potential threats. The support team will provide technical assistance to barangay staff whenever issues will occur, such as login difficulties, system errors, or problems with data retrieval. This immediate response will help minimize downtime and will ensure uninterrupted access to household records.

In addition to routine corrective maintenance, adaptive maintenance will be implemented to accommodate changes in the operational environment. For example, adjustments will be made to ensure compatibility with updated software platforms, web browsers, or operating systems. Enhancements suggested during the review phase, such as improved reporting formats and stronger security policies,

will gradually integrate into the system. These refinements not only will improve functionality but will also reinforce user confidence in the system's reliability.

The support aspect of this phase will also include continuous training and guidance for barangay officials and staff. As new features will be introduced or improvements will be made, users will be given orientation sessions to keep them updated with system changes. This will ensure that the system will remain easy to use, even as it evolved. Moreover, documentation and user manuals will regularly revise to serve as practical guides for troubleshooting and performing basic operations.

Periodic system audits will be also conducted to assess data accuracy, check for inconsistencies, and verify compliance with security measures. Backup and recovery protocols will be regularly tested to safeguard information in case of system failure or unforeseen incidents such as power interruptions or connectivity issues. By consistently monitoring these aspects, the researchers ensures that household records will always remain secure and accessible.

Ultimately, the maintenance and support phase will highlight the importance of sustainability in system development. The Barangay-Connect system will not be viewed as a one-time solution but as a dynamic tool that will require continuous care, updates, and improvements. By providing consistent technical support, incorporating user feedback, and adapting to technological changes, the system will be able to maintain its effectiveness in improving barangay household profiling. This phase will ensure that the project will remain beneficial to the

community of San Isidro, Nueva Ecija, while establishing a model that could potentially be adopted by other barangays in the future.

2. Assessment of the Technical Quality of the Barangay-Connect by IT Experts based on the ISO 25010 standards.

The assessment conducted by IT experts on the Barangay-Connect: Barangay Household Profiling System++ shows that the system performed excellently in all major software quality parameters based on the ISO 25010 standards. The results from the different tables indicate that the system meets professional standards for software functionality, performance, compatibility, usability, reliability, security, maintainability, and portability.

2.1. Functional Suitability

Table 10 shows the results of assessment made by the IT Experts on the functional suitability of the developed Barangay-Connect.

Table 11. Results of IT Experts' Assessment on the Application's Functional Suitability

Criteria	Functional Suitability	
	Weighted Mean	Qualitative Rating
Functional Completeness	4.80	Excellent
Functional Correctness	4.60	Excellent
Functional Appropriateness	4.40	Excellent
Grand Mean	4.60	Excellent

In general, **Functional Suitability** of the Barangay-Connect obtained a grand mean of **4.60**, which is qualitatively rated as **Excellent**. The sub-criteria under this parameter include functional completeness with a mean of 4.80, functional correctness at 4.60, and functional appropriateness at 4.40. These results imply that the system successfully delivers all required features, performs its functions accurately, and provides outputs that align well with the intended purpose. The high scores suggest that the system effectively meets user needs and expectations, with minimal errors or missing components.

2.2. Performance Efficiency

Table 11 shows how the IT Experts assessed the Barangay-Connect on performance efficiency.

Table 12: Results of IT Experts' Assessment on the Application's Performance Efficiency

Criteria	Performance Efficiency	
	Weighted Mean	Qualitative Rating
Time Behavior	4.80	Excellent
Resource Utilization	4.60	Excellent
Grand Mean	4.70	Excellent

For **Performance Efficiency**, the grand mean was **4.70**, also rated as **Excellent**. The specific parameters of time behavior (4.80) and resource utilization (4.60) both received high ratings, reflecting that the system can process data efficiently and operating smoothly without overusing system resources. This

suggests that the application is optimized for performance and can handle multiple tasks or users simultaneously without significant delays or system lags.

2.3. Compatibility

Table 12 shows the assessment result made by the IT Experts on the compatibility of Barangay-Connect.

Table 13: Results of IT Experts' Assessment on the Application's Compatibility

Criteria	Compatibility	
	Weighted Mean	Qualitative Rating
Co-Existence	4.80	Excellent
Resource Utilization	4.80	Excellent
Grand Mean	4.80	Excellent

The system's **Compatibility** achieved an impressive grand mean of **4.80**, which was rated **Excellent**. Both co-existence and resource utilization received the same weighted mean of 4.80. This indicates that Barangay-Connect can function effectively alongside other systems and hardware without conflicts or errors. The results demonstrate that the system is highly adaptable and can be integrated within existing technological infrastructures used by barangays or local government offices.

2.4. Usability

Table 13 shows the results of assessment made by the IT Experts on the usability of Barangay-Connect.

Table 14: Results of IT Experts' Assessment on the Application's Usability

Criteria	Usability	
	Weighted Mean	Qualitative Rating
Appropriateness Recognizability	4.80	Excellent
Learnability	4.80	Excellent
Operability	5.00	Excellent
User Error Protection	4.00	Very Good
User Interface Aesthetics	4.40	Excellent
Grand Mean	4.60	Excellent

In the area of **Usability**, the system obtained a grand mean of **4.60**, again rated **Excellent**. Appropriateness recognizability and learnability both scored 4.80, while operability reached the highest possible mean of 5.00. However, user error protection (4.00) and user interface aesthetics (4.40) received slightly lower ratings, which, though still excellent, suggest minor areas for improvement. These results imply that while the system is easy to learn and operate, it could benefit from enhancements in visual design and stronger measures to prevent user input errors. Overall, the findings show that users can easily understand, navigate, and operate the system with minimal training.

2.5. Reliability

Table 14 shows the IT Experts assessment on the reliability of the Barangay-Connect.

Table 15: Results of IT Experts' Assessment on the Application's Reliability

Criteria	Reliability	
	Weighted Mean	Qualitative Rating
Maturity	4.80	Excellent
Availability	4.80	Excellent
Recoverability	4.60	Excellent
Grand Mean	4.70	Excellent

Regarding **Reliability**, Barangay-Connect achieved a grand mean of **4.70**, rated **Excellent**. The sub-criteria—maturity and availability (4.80 each) and recoverability (4.60)—indicate that the system performs consistently well and can recover effectively from potential disruptions or failures. This confirms that Barangay-Connect is stable and dependable for day-to-day use, capable of maintaining consistent performance without compromising data integrity.

2.6. Security

Table 15 shows how the IT Experts assesses the Barangay-Connect on security.

Table 16: Results of IT Experts' Assessment on the Application's Security

Criteria	Security	
	Weighted Mean	Qualitative Rating
Confidentiality	4.00	Excellent
Integrity	4.60	Excellent
Grand Mean	4.30	Excellent

The **Security** parameter, though still rated **Excellent**, recorded a slightly lower grand mean of **4.30**. The confidentiality criterion received a rating of 4.00, interpreted as **Very Good**, while integrity was rated 4.60. This means that while the system protects data from unauthorized changes and maintains accuracy, there is room for improvement in ensuring stricter confidentiality and access control mechanisms. Enhancing data encryption, authentication procedures, and privacy safeguards would strengthen this aspect further.

2.7. Maintainability

Table 16 shows the assessment result of the Barangay-Connect on maintainability.

Table 17: Results of IT Experts' Assessment on the Application's Maintainability

Criteria	Maintainability	
	Weighted Mean	Qualitative Rating
Modularity	4.80	Excellent
Reusability	4.80	Excellent
Analyzability	5.00	Excellent
Modifiability	4.80	Excellent
Testability	4.80	Excellent
Grand Mean	4.84	Excellent

When it comes to **Maintainability**, Barangay-Connect achieved the highest score among all parameters, with a grand mean of **4.84**, rated **Excellent**. Modularity, reusability, modifiability, and testability all scored 4.80, while

analyzability reached a perfect score of 5.00. These results imply that the system is well-designed in terms of structure and documentation, making it easy for developers or administrators to analyze, modify, and update when needed. It also suggests that the system's codebase is modular and reusable, which facilitates long-term sustainability and efficient system evolution.

2.8. Portability

Table 17 shows the IT Experts assessment of the Barangay-Connect on portability.

Table 18: Results of IT Experts' Assessment on the Application's Portability

Criteria	Portability	
	Weighted Mean	Qualitative Rating
Adaptability	4.80	Excellent
Installability	4.80	Excellent
Replaceability	4.80	Excellent
Grand Mean	4.80	Excellent

In **Portability**, the system earned a grand mean of **4.80**, rated **Excellent**. The parameters of adaptability, installability, and replaceability each scored 4.80, reflecting that the system can be easily installed, configured, and deployed in different environments. This high rating indicates that Barangay-Connect can be effectively replicated and utilized in other barangays or local government settings without major compatibility issues.

Table 19. Summary of Assessment of IT Experts on the Technical Quality of the Barangay-Connect based on the ISO 25010 Software Quality Standards

Technical Quality	Quantitative Rating	Qualitative Rating
Functional Suitability	4.60	Excellent
Performance Efficiency	4.70	Excellent
Compatibility	4.80	Excellent
Usability	4.40	Excellent
Reliability	4.60	Excellent
Security	4.30	Excellent
Maintainability	4.84	Excellent
Portability	4.80	Excellent
Grand Mean	4.63	Excellent

Overall, the results reveal that Barangay-Connect is a highly efficient, functional, and dependable system that meets professional standards in software quality. The consistent “Excellent” ratings across nearly all categories confirm that the system is well-developed, user-friendly, and technically sound. While the security and user interface could benefit from minor enhancements, the overall performance demonstrates that Barangay-Connect is ready for full implementation and can effectively support local governance and household profiling operations in San Isidro, Nueva Ecija.

3. Assessment on the use of the application by the barangay staff

The assessment results provided by the barangay staff on the Barangay-Connect: Barangay Household Profiling System++ reveal an overall outstanding evaluation across all software quality parameters. The uniformly high weighted means and consistent

qualitative ratings indicate that the system has successfully met, and in many aspects exceeded, the expectations of its end-users at the barangay level.

3.1. Functional Suitability

Table 19 shows the result of the assessment made by the barangay staff on the Barangay-Connect in terms of functional suitability.

Table 20. Results of Barangay Staff's Assessment on the Application's Functional Suitability

Criteria	Functional Suitability	
	Weighted Mean	Qualitative Rating
Functional Completeness	5.00	Excellent
Functional Correctness	5.00	Excellent
Functional Appropriateness	5.00	Excellent
Grand Mean	5.00	Excellent

In the area of **Functional Suitability**, Barangay-Connect received perfect scores across all sub-criteria, resulting in a grand mean of **5.00**, which is interpreted as **Excellent**. This parameter measures how well the system meets its intended purpose and provides the necessary functionalities required for barangay operations. The perfect ratings for functional completeness, correctness, and appropriateness imply that the system offers all the essential features expected by users, produces accurate outputs, and performs its functions effectively in accordance with the needs of the community. Such a result demonstrates that the system was able to align seamlessly with the tasks of profiling and

managing household data, confirming its reliability and comprehensiveness from a user perspective.

3.2. Usability

Table 20 shows the results of assessment made by the Barangay Staff on the usability of Barangay-Connect.

Table 21: Results of Barangay Staff's Assessment on the Application's Usability

Criteria	Usability	
	Weighted Mean	Qualitative Rating
Appropriateness Recognizability	5.00	Excellent
Learnability	5.00	Excellent
Operability	5.00	Excellent
User Error Protection	5.00	Excellent
User Interface Aesthetics	5.00	Excellent
Grand Mean	5.00	Excellent

For **Usability**, the assessment also resulted in a grand mean of **5.00**, rated as **Excellent**. The barangay staff rated every aspect of usability, including appropriateness recognizability, learnability, operability, user error protection, and user interface aesthetics, with perfect scores. This means that users found the system extremely easy to learn, navigate, and operate, even for those with limited technical background. The findings suggest that Barangay-Connect provides an intuitive and user-friendly interface that enhances user satisfaction and promotes efficiency. Additionally, the high score in user error protection indicates that the

system effectively minimizes input mistakes and provides clear prompts or feedback to guide users. The uniform perfect rating across all usability aspects signifies that the design, layout, and overall interaction experience of Barangay-Connect are exceptionally well-crafted and meet user expectations.

3.3. Reliability

Table 21 shows the Barangay Staff assessment on the reliability of the Barangay-Connect.

Table 22: Results of Barangay Staff's Assessment on the Application's Reliability

Criteria	Reliability	
	Weighted Mean	Qualitative Rating
Maturity	5.00	Excellent
Availability	5.00	Excellent
Recoverability	5.00	Excellent
Grand Mean	5.00	Excellent

In terms of **Reliability**, the system again obtained a grand mean of **5.00**, which corresponds to an **Excellent** qualitative rating. Reliability pertains to the system's ability to consistently perform as expected without failure. The perfect scores in maturity, availability, and recoverability confirm that Barangay-Connect operates with stability and dependability. This means that the system remains consistently functional under different conditions, is readily available for use without frequent downtime, and can recover smoothly from potential errors or interruptions. Such results reflect a strong and well-tested system infrastructure,

ensuring that users can depend on it for continuous and accurate data management at the barangay level.

3.4. Security

Table 22 shows how the Barangay Staff assesses the BARANGAY-CONNECT on security.

Table 23: Results of Barangay Staff's Assessment on the Application's Security

Criteria	Security	
	Weighted Mean	Qualitative Rating
Confidentiality	5.00	Excellent
Integrity	5.00	Excellent
Grand Mean	5.00	Excellent

The evaluation of **Security** also yielded a grand mean of **5.00**, interpreted as **Excellent**. Both confidentiality and integrity scored 5.00, indicating that users perceive the system as highly secure and capable of protecting sensitive data. The rating suggests that Barangay-Connect implements effective mechanisms to ensure that only authorized personnel have access to data, and that stored information remains unaltered and protected from potential breaches. Although security is often a critical concern in community-level information systems, the perfect ratings reflect strong confidence from the barangay staff regarding the system's capability to safeguard private household records and ensure ethical data management practices.

Table 24. Summary of Assessment of the Barangay Staff on the Quality of Use of the Barangay-Connect based on the ISO 25010 Software Quality Standards

Technical Quality	Quantitative Rating	Qualitative Rating
Functional Suitability	5.00	Excellent
Usability	5.00	Excellent
Reliability	5.00	Excellent
Security	5.00	Excellent
Grand Mean	5.00	Excellent

Overall, the barangay staff's assessment demonstrates a unanimous and highly positive perception of Barangay-Connect's performance and quality. With all parameters achieving perfect grand means of **5.00** and qualitative ratings of **Excellent**, it is evident that the system met every major software quality standard from the perspective of its intended users. This outcome indicates that Barangay-Connect is not only technically sound but also highly functional, secure, and user-friendly in its actual deployment context. The findings affirm the system's readiness for full-scale implementation and its significant potential to improve local governance efficiency and accuracy in household profiling operations across barangays.

4. Assessment on the use of the application by the head of households

The overall evaluation of the Barangay-Connect application reveals outstanding performance across all assessed parameters: usability, reliability, and security.

4.1. Usability

Table 24 shows the results of assessment made by the Heads of Household on the usability of Barangay-Connect.

Table 25: Results of Heads of Household's Assessment on the Application's Usability

Criteria	Usability	
	Weighted Mean	Qualitative Rating
Appropriateness Recognizability	5.00	Excellent
Learnability	4.80	Excellent
Operability	4.50	Excellent
User Error Protection	5.00	Excellent
User Interface Aesthetics	4.80	Excellent
Grand Mean	4.82	Excellent

For **Usability**, the results show consistently high ratings, with all criteria assessed as **Excellent**. Appropriateness and recognizability, as well as user error protection, both achieved perfect weighted means of 5.00, indicating that the system is well-designed for its intended purpose and effectively minimizes user mistakes. Learnability and user interface aesthetics both received a 4.80 rating, showing that users find the application intuitive and visually appealing. Operability, rated 4.50, remains excellent but suggests minor room for improvement in terms of user interaction and task execution. The overall grand mean of **4.82** confirms that the application is perceived as highly usable by its users.

4.2. Reliability

Table 25 shows the Head of Households' assessment on the reliability of the Barangay-Connect.

Table 26: Results of Heads of Household's Assessment on the Application's Reliability

Criteria	Reliability	
	Weighted Mean	Qualitative Rating
Maturity	5.00	Excellent
Availability	5.00	Excellent
Recoverability	5.00	Excellent
Grand Mean	5.00	Excellent

Regarding **Reliability**, the application achieved perfect ratings across all sub-criteria. Maturity, availability, and recoverability each obtained a weighted mean of 5.00 and were all rated **Excellent**. This indicates that the system performs consistently, remains accessible to users, and can recover efficiently from any operational disruptions. The grand mean of **5.00** reinforces that the application demonstrates exceptional dependability and stability.

4.3. Security

Table 26 shows how the Heads of Household assesses the Barangay-Connect on security.

Table 27: Results of Heads of Household's Assessment on the Application's Security

Criteria	Security	
	Weighted Mean	Qualitative Rating
Confidentiality	5.00	Excellent
Integrity	5.00	Excellent
Grand Mean	5.00	Excellent

In the parameter of **Security**, both confidentiality and integrity were rated highly, with each receiving a weighted mean of 5.00. Confidentiality and integrity was both rated **Excellent**. This suggests that the application ensures data integrity and protection, though there is a minor perception that confidentiality could still be slightly enhanced. The grand mean of **5.00** nonetheless indicates that overall, users view the system as secure and trustworthy.

Table 28. Summary of Assessment of the Heads of Household on the Quality of Use of the Barangay-Connect based on the ISO 25010 Software Quality Standards

Technical Quality	Quantitative Rating	Qualitative Rating
Usability	4.82	Excellent
Reliability	5.00	Excellent
Security	5.00	Excellent
Grand Mean	4.94	Excellent

In summary, the assessment results demonstrate that Barangay-Connect excels in usability, reliability, and security. With consistent **Excellent** ratings and a perfect grand mean across all parameters, the system is validated as a highly effective, stable, and secure application, well-received by the Head of Households who participated in the evaluation.

Chapter IV

Summary of Findings, Conclusion, and Recommendation

This chapter provides a comprehensive summary of the entire research study, highlighting its key objectives, methodology, and major findings. It also presents the conclusions drawn from the results, which are directly aligned with the research questions and objectives. Furthermore, this chapter offers practical recommendations based on the study's outcomes, as well as suggestions for future research that may further enhance understanding and address the limitations encountered in this study.

Summary

The study entitled “*Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija*” was conducted to design and develop a digital solution that streamlines the process of household profiling and information management at the barangay level. Traditionally, barangay offices rely on manual data collection and record-keeping methods, which are often time-consuming, error-prone, and inefficient. To address these challenges, the researchers developed Barangay-Connect, a computerized system designed to simplify data gathering, storage, retrieval, and updating of household information.

The system aims to assist barangay officials in efficiently maintaining up-to-date records of residents, generating accurate reports, and improving decision-making processes related to community planning, resource allocation, and public service delivery. Using modern software development methodologies, the system was designed with features for user management, data entry, record searching, report generation, and data security.

The study employed both developmental and descriptive research methods. Evaluation of the system was carried out by IT experts and selected heads of households based on several parameters, including usability, reliability, and security. The assessment results revealed outstanding performance across all areas, with grand means consistently rated as *Excellent*. Respondents found the system highly usable, reliable, and secure, demonstrating that Barangay-Connect meets the functional and qualitative standards expected from a modern information management system.

The Barangay-Connect system successfully addressed the limitations of the manual profiling process and provided an efficient, accurate, and secure digital platform for barangay operations. The study recommends further enhancement of the system to include integration with municipal databases and mobile access to improve accessibility and scalability for wider community use.

Findings of the Study

1. Development of BARANGAY-CONNECT: Barangay Household Profiling System++ for San Isidro, Nueva Ecija.

The development of Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija followed the phases of the Software Development Life Cycle (SDLC), which ensured a systematic and organized approach in designing, developing, and evaluating the system. Each phase contributed to the creation of a fully functional and efficient system that meets the needs of barangay officials in managing household records.

The findings revealed that the Barangay-Connect system successfully met its objectives of automating and improving the household profiling process in San Isidro, Nueva Ecija. The integration of the SDLC phases ensured that the system was systematically developed, functionally robust, and highly satisfactory to its intended users.

2. Assessment of the Technical Quality of the Barangay-Connect by IT Experts based on the ISO 25010 standards.

The findings of the study, based on the assessment conducted by IT experts using the ISO 25010 Software Quality Standards, revealed that Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija performed excellently across all major software quality parameters.

The results affirm that Barangay-Connect is a highly functional, efficient, and dependable system that adheres to international software quality standards. The consistently excellent ratings across all parameters confirm that the system is technically sound, user-friendly, and ready for implementation to support effective and streamlined household profiling and local governance in San Isidro, Nueva Ecija.

3. Assessment on the use of the application by the barangay staff

The findings of the study revealed that Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija obtained an overall **Excellent** across all parameters based on the ISO 25010 Software Quality

Standards. The barangay staff assessed the system's **Functional Suitability, Usability, Reliability, and Security**, all of which achieved perfect scores. This indicates that the system effectively performs all required functions, operates efficiently, and provides accurate and secure management of household data. Users found Barangay-Connect highly intuitive, easy to navigate, and dependable, with strong data protection and error prevention features. These findings demonstrate that the system has successfully met and even exceeded user expectations, proving it to be a highly functional, secure, and user-friendly tool for improving barangay operations and household profiling in San Isidro, Nueva Ecija.

4. Assessment on the use of the application by the heads of household

The findings of the study revealed that Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija obtained an overall qualitative rate as Excellent, based on the assessment made by the Heads of Households following the ISO 25010 Software Quality Standards. The system performed excellently in all parameters—Usability, Reliability, and Security. These results demonstrate that Barangay-Connect is a highly functional, dependable, and user-friendly system that effectively supports efficient and secure household profiling operations in San Isidro, Nueva Ecija.

Conclusion

Based on the findings, the following conclusions were drawn:

1. Guided by the Software Development Life Cycle (SDLC), the development process followed a structured and systematic approach, ensuring that each phase—from planning to deployment—contributed to the creation of a fully functional and reliable system tailored to the needs of barangay officials and residents.

The development process successfully produced a system that automated the manual profiling procedures previously used in San Isidro, Nueva Ecija. Findings confirmed that the Barangay-Connect system effectively met its primary objectives by improving record management, minimizing human error, and providing timely and accurate household data for decision-making. The integration of SDLC principles ensured that the system was developed logically, thoroughly tested, and optimized for user satisfaction and operational performance.

2. The assessment of the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija by IT experts using the ISO 25010 Software Quality Standards confirms that the system possesses outstanding technical quality and adheres to international software development benchmarks.

These results affirm that Barangay-Connect is a technically sound, efficient, and dependable system that meets the highest standards of software quality. The consistently excellent ratings across all ISO 25010 parameters validate its robustness, scalability, and readiness for deployment. Barangay-Connect stands as a model of technological innovation in local governance—capable of streamlining

household profiling operations, improving data accuracy, and supporting informed decision-making in San Isidro, Nueva Ecija. Its exemplary performance demonstrates strong potential for adoption and replication in other communities seeking to modernize their data management systems.

3. The findings of the study confirm that Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija has fully achieved its purpose of enhancing the efficiency, accuracy, and security of household data management at the barangay level. Based on the evaluation conducted by barangay staff using the ISO 25010 Software Quality Standards, the system obtained a qualitatively rating of Excellent across all evaluated parameters.

The results indicate that Barangay-Connect has successfully met and even exceeded user expectations. It stands as a highly functional, secure, and user-friendly tool that supports efficient household profiling and strengthens barangay governance. The system's excellent performance proves its readiness for full implementation and its potential to serve as a model for other barangays aiming to adopt digital solutions for effective community information management.

4. The assessment conducted among the heads of households regarding the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija yielded highly positive results, affirming that Barangay-Connect is a highly functional, secure, and user-friendly system that effectively supports efficient and trustworthy household profiling operations in San Isidro, Nueva Ecija. The

excellent evaluation results from the heads of households validate that the system not only fulfills its intended purpose but also enhances community engagement by providing a reliable and easy-to-use platform for data management.

The outstanding performance of Barangay-Connect from the perspective of household users reinforces its potential as a valuable digital innovation for local governance. It stands as a testament to how well-designed information systems can empower communities, improve service delivery, and strengthen the accuracy and transparency of barangay data management.

Recommendations

Based on the findings and conclusions of the study, several recommendations are proposed to ensure the continuous improvement, sustainability, and broader applicability of Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija. These recommendations are directed to the application, barangay officials, end users, and future researchers.

- 1. Implementation and Adoption.** It is highly recommended that the local government of San Isidro, Nueva Ecija formally adopt the Barangay-Connect system as its official platform for household profiling and record management. The system's excellent technical performance and user satisfaction indicate that it can significantly enhance the efficiency, accuracy, and transparency of barangay operations. Full implementation across all barangays in San Isidro will streamline data collection, reduce manual record-keeping, and promote data-driven governance.

2. **Continuous System Maintenance and Updates.** To sustain the system's effectiveness, regular maintenance and updates should be conducted. The system must undergo continuous monitor for technical issues, security vulnerabilities, and performance inefficiencies. Periodic updates should also include improvements in user interface design, database optimization, and integration with new technologies to maintain high performance standards. Establishing a dedicated technical support team is recommended to ensure immediate troubleshooting and user assistance.
3. **Enhancement of Security and Privacy Features.** Although the system performed excellently around security, continuous enhancement of data protection mechanisms is essential to safeguard sensitive household information. The system should integrate advanced encryption techniques, secure authentication protocols, and role-based access control to further strengthen privacy and data integrity. Implementing an automated data backup and disaster recovery system is also advisable to ensure data availability and reliability in case of system failures.
4. **User Training and Capacity Building.** Comprehensive training programs should be organized for barangay officials and staff who will utilize the system. These training sessions should focus on proper data entry, system navigation, and maintenance procedures to minimize user errors and ensure data accuracy. Refresher courses and user manuals should be made available to maintain consistent proficiency and to support new personnel who will use the system in the future.
5. **Expansion and System Integration.** To maximize the system's potential, future developments should aim to expand its features and capabilities. The system could

be integrated with municipal and provincial databases to allow real-time data sharing and collaboration between local government units (LGUs). Additional functionalities, such as population analytics, health monitoring, and disaster response profiling, could also be incorporated to enhance its value in governance and community planning.

6. **Replication and Scalability.** Given the system's excellent technical quality and positive user evaluations, Barangay-Connect is highly recommended for replication in other barangays within Nueva Ecija and neighboring municipalities. Its modular and portable design makes it adaptable to different barangay environments with minimal customization. Replicating the system can contribute to a unified, province-wide digital database that will improve the overall efficiency of local governance and public service delivery.
7. **Future Research and Development.** Future researchers are encouraged to explore enhancements that incorporate emerging technologies such as cloud computing, mobile applications, artificial intelligence, and data analytics. These innovations could further improve the accessibility, scalability, and analytical capacity of Barangay-Connect. Moreover, longitudinal studies may be conducted to assess the long-term impact of the system on barangay efficiency, transparency, and citizen satisfaction.

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APPENDIX A
Letter of Permission to Conduct a Study

	Republic of the Philippines NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY Cabanatuan City, Nueva Ecija, Philippines ISO 9001:2015 CERTIFIED									
SAN ISIDRO CAMPUS – College of Information and Communications Technology										
February 14, 2025										
Hon. Zenaida Del Rosario Eugenio Barangay Captain Poblacion, San Isidro Nueva Ecija Dear Madam, Greetings! We, the Bachelor of Science in Information Technology students of Nueva Ecija University of Science and Technology are presently in the preliminary stages of writing our project charter entitled "Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija " as part of our subject in IT-CAP 01 Capstone Project and Research 1. Due to this respect, may we ask your permission to allow us to conduct online/face to face interviews with you and your staff and observe area operations as part of Phase 1 Preliminary Investigation? May we also request for sample documents or reports relevant to the operation. The documents will serve as basis in determining input and output information. We assure you that all information gathered will be held strictly confidential and for research purpose only. Very Truly Yours, <table style="width: 100%; margin-top: 20px;"><tr><td style="width: 50%; text-align: center;">MR. ART D. ANGELES</td><td style="width: 50%; text-align: center;">MS. AVEGAIL V. PANGILINAN</td></tr><tr><td style="text-align: center;">MR. RICKY JR. P. DELA CRUZ</td><td style="text-align: center;">MR. LEMUEL M. REBULTAN</td></tr><tr><td style="text-align: center;">MR. RJ G. SOTTO</td><td></td></tr></table> Recommending Approval: <table style="width: 100%; margin-top: 20px;"><tr><td style="width: 50%; text-align: center;">ARLIE C. SANTOS, MSIT Capstone Project and Research 1 Instructor</td><td style="width: 50%; text-align: center;">MARVIN DG. GARCIA, MSIT CICT Chairperson</td></tr></table> Approved: Hon. Zenaida Del Rosario Eugenio Barangay Captain			MR. ART D. ANGELES	MS. AVEGAIL V. PANGILINAN	MR. RICKY JR. P. DELA CRUZ	MR. LEMUEL M. REBULTAN	MR. RJ G. SOTTO		ARLIE C. SANTOS, MSIT Capstone Project and Research 1 Instructor	MARVIN DG. GARCIA, MSIT CICT Chairperson
MR. ART D. ANGELES	MS. AVEGAIL V. PANGILINAN									
MR. RICKY JR. P. DELA CRUZ	MR. LEMUEL M. REBULTAN									
MR. RJ G. SOTTO										
ARLIE C. SANTOS, MSIT Capstone Project and Research 1 Instructor	MARVIN DG. GARCIA, MSIT CICT Chairperson									
<i>Transforming Communities through Science and Technology</i> <table style="width: 100%;"><tr><td style="width: 60%;"></td><td style="width: 40%; text-align: right;"> Contact No. (0 44) 940-9980 Email: sanidro@neust.edu.ph www.neust.edu.ph </td></tr></table>				Contact No. (0 44) 940-9980 Email: sanidro@neust.edu.ph www.neust.edu.ph						
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SAN ISIDRO CAMPUS – College of Information and Communications Technology

Bachelor of Science in Information Technology

Web Systems Technology (WST) Major

RESEARCHERS' CONTACT DETAILS

NO	STUDENT'S NAME (picture & Name)	CONTACT NUMBER	EMAIL ADDRESS	FB/MESSENGER NAME
1	 ART D. ANGELES	09532579371	angelesart28@gmail.com	Art D. Angeles
2	 RICKY JR. P. DELA CRUZ	09464290517	sairidpineda28@gmail.com	Ricky P. Dela Cruz
3	 RJ G. SOTTO	09552800990	sotto.rj01@gmail.com	Rj Gregorio Sotto
4	 AVEGAIL V. PANGILINAN	09755122949	iamavegailpangilinan@gmail.com	Avegail Pangilinan

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5	 LEMUEL M. REBULTAN	09676209021	rebultanlemuel@gmail.com	Rebultan M. Limuel
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SAN ISIDRO CAMPUS – College of Information and Communications Technology

INTERVIEW QUESTIONS

Date:	Time:
Interviewer:	Interviewee:
Objectives of the interview	
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INTERVIEWER'S QUESTIONS	INTERVIEWEE'S ANSWERS
1. How do you gather data from your residents? What information do you get from them? Do you have templates/datasets coming from the LGU on what kind of data you need to get?	
2. Where do you keep data coming from residents? How often you get data from the residents?	
3. What is the usual volume of data flow from your data gathering? Aside from yourself, who usually get the data from the residents?	
4. On the average, how many residents ask for certificates in a day? Who usually prepare the certificates? How long will it take the certificate to prepare?	
5. What kind of reports do you submit to the LGU? How often do they require you to submit? How long does it take for you to prepare the reports?	



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SAN ISIDRO CAMPUS – College of Information and Communications Technology

February 14, 2025

Hon. Zenaida Del Rosario Eugenio
Barangay Captain
Poblacion, San Isidro
Nueva Ecija

Dear Madam,

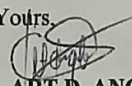
Greetings!

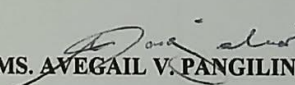
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Due to this respect, may we ask your permission to allow us to conduct **online/face to face interviews** with you and your staff and observe area operations as part of Phase 1 Preliminary Investigation? May we also request for **sample documents** or reports relevant to the operation. The documents will serve as basis in determining input and output information.

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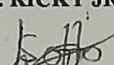
Very Truly Yours,


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MS. AVEGAIL V. PANGILINAN


MR. RICKY JR. P. DELA CRUZ


MR. LEMUEL M. REBULTAN

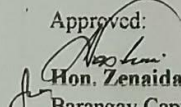

MR. RJ C. SOTTO

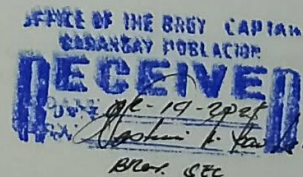
Recommending Approval:


ARLIE C. SANTOS, MSIT
Capstone Project and Research 1 Instructor


MARVIN DG. GARCIA, MSIT
CICT Chairperson

Approved:


Hon. Zenaida Del Rosario Eugenio
Barangay Captain





Republic of the Philippines

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SAN ISIDRO CAMPUS – College of Information and Communications Technology

February 14, 2025

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Municipal Administrator
Municipality of San Isidro
Nueva Ecija

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Approved:

MS. MARY GRACE T. CARIÑO
Municipal Administrator



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February 14, 2025

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Municipality of San Isidro
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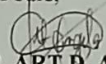
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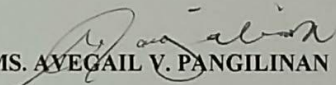
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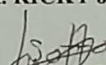
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MR. ART D. ANGELES


MS. AYE GAIL V. PANGILINAN

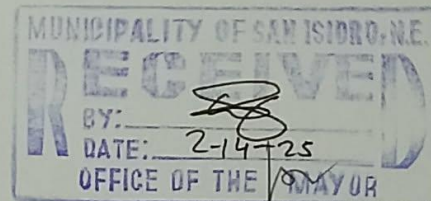

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MR. LEMUEL M. REBULTAN


MR. RJ G. SOTTO

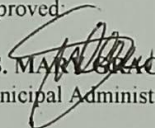
Recommending Approval:


ARLIE C. SANTOS, MSIT
Capstone Project and Research 1 Instructor



MARVIN DG. GARCIA, MSIT
CICT Chairperson

Approved:


MS. MARY GRACE T. CARIÑO
Municipal Administrator

APPENDIX B
Letter of Permission to Conduct Validation



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

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SAN ISIDRO CAMPUS – College of Information and Communications Technology

Hon. Zenaida Del Rosario Eugenio
Barangay Captain
Poblacion, San Isidro

Bachelor of Science in Information Technology Students
Nueva Ecija University of Science and Technology

Dear Madam,

We are pleased to introduce **Barangay-Connect: Barangay Household Profiling System++**, a project designed to modernize the way the barangay collects, organizes, and maintains household data while strengthening residents' access to healthcare services.

Managing resident records, particularly those related to health, can be a challenge for barangay workers. Our goal with **Barangay-Connect** is to reduce the administrative burden on barangay staff by providing an efficient digital system for household profiling, while also improving healthcare service delivery for residents. This system will allow the barangay to maintain accurate and up-to-date household records, ensuring that resident information is well-organized and easily accessible for better planning and decision-making.

Beyond data management, **Barangay-Connect** will serve as a bridge between barangay residents and the Rural Health Unit. Through the system, residents will be able to schedule medical appointments, minimizing long queues and improving the flow of patients at the health center. Additionally, the platform will provide residents with access to their health records, allowing them to track their medical history and previous consultations with ease.

The primary objectives of **Barangay-Connect** are to streamline the barangay's household data management, improve coordination with the RHU, and enhance accessibility to essential healthcare services. By achieving these goals, the system aims to simplify administrative tasks, provide better healthcare support to residents, and promote a more efficient and connected barangay.

We are excited about the potential impact of this project and appreciate your support and collaboration in making it a reality for Barangay San Isidro.

Sincerely,

Art D. Angeles

Ricky Jr. P. Dela Cruz

Avegail V. Pangilinan

Rj G. Sotto

Lemuel M. Rebutan

Contact Information: 09532579371



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

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Barangay Captain

Poblacion, San Isidro

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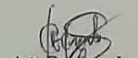
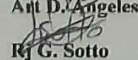
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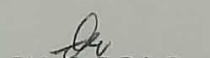
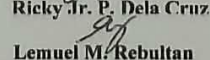
Managing resident records, particularly those related to health, can be a challenge for barangay workers. Our goal with Barangay-Connect is to reduce the administrative burden on barangay staff by providing an efficient digital system for household profiling, while also improving healthcare service delivery for residents. This system will allow the barangay to maintain accurate and up-to-date household records, ensuring that resident information is well-organized and easily accessible for better planning and decision-making.

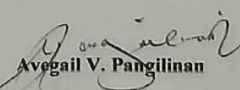
Beyond data management, **Barangay-Connect** will serve as a bridge between barangay residents and the Rural Health Unit. Through the system, residents will be able to schedule medical appointments, minimizing long queues and improving the flow of patients at the health center. Additionally, the platform will provide residents with access to their health records, allowing them to track their medical history and previous consultations with ease.

The primary objectives of **Barangay-Connect** are to streamline the barangay's household data management, improve coordination with the RHU, and enhance accessibility to essential healthcare services. By achieving these goals, the system aims to simplify administrative tasks, provide better healthcare support to residents, and promote a more efficient and connected barangay.

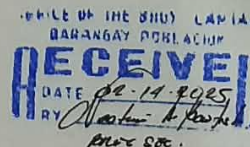
We are excited about the potential impact of this project and appreciate your support and collaboration in making it a reality for Barangay San Isidro. Sincerely,


Art D. Angeles

Rj G. Sotto


Ricky Jr. P. Dela Cruz

Lemuel M. Rebultan


Avegail V. Pangilinan

Contact Information: 09532579371





Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

Ms. Mary Grace T. Cariño

Municipality Administrator

Municipal of San Isidro

Bachelor of Science in Information Technology Students

Nueva Ecija University of Science and Technology

Dear Madam,

We are pleased to introduce **Barangay-Connect: Barangay Household Profiling System++**, a project designed to modernize the way the barangay collects, organizes, and maintains household data while strengthening residents' access to healthcare services.

Managing resident records, particularly those related to health, can be a challenge for barangay workers. Our goal with **Barangay-Connect** is to reduce the administrative burden on barangay staff by providing an efficient digital system for household profiling, while also improving healthcare service delivery for residents. This system will allow the barangay to maintain accurate and up-to-date household records, ensuring that resident information is well-organized and easily accessible for better planning and decision-making.

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Avegail V. Pangilinan

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Republic of the Philippines

NUEVA ECIIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

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Municipality Administrator

Municipal of San Isidro

Bachelor of Science in Information Technology Students

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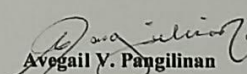
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Ar D. Angeles

R/G Sotto


Ricky Jr. P. Dela Cruz

Lemuel M. Rebultan


Avegail Y. Pangilinan

Contact Information: 09532579371

APPENDIX C
Letter to IT Experts



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

September 16, 2025

MRS. MARY ANN L. DE GUZMAN
Teacher III
T.A. Dionisio National High School
Malapit, San Isidro, Nueva Ecija

Ma'am,

Greetings!

We, the Bachelor of Science in Information Technology students, are in the final stage of stages of writing our Capstone Project entitled "Barangay-Connect: Barangay Household Profiling System ++ for San Isidro, Nueva Ecija "

May we have the honor to request permission to use your office as our respondent for this project? Requesting further that we be allowed to validate the system through the assistance and answers of your staff. Feedback and comments will be used for modification and improving the system.

Thank you very much and looking forward for your approval.

Respectfully yours,

ART D. ANGELES

RJ G. SOTTO

RICKY JR. P. DELA CRUZ

AVEGAIL V. PANGILINAN

LEMUEL M. REBULTAN

Noted:

Recommending Approval:

MRS. ARLIE C. SANTOS, MSIT
Capstone Project and Research I Instructor

MR. MARVIN D.G. GARCIA, MSIT
CICT Chairperson

Conforme:

MRS. MARY ANN L. DE GUZMAN
IT Expert



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

September 16, 2025

MRS. MARY ANN L. DE GUZMAN
Teacher III
T.A. Dionisio National High School
Malapit, San Isidro, Nueva Ecija

Ma'am,

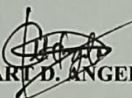
Greetings!

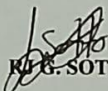
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RICKY JR. P. DELA CRUZ


AVEGAIL V. PANGILINAN


LEMUEL M. REBULTAN

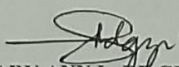
Noted:


MRS. ARLIE C. SANTOS, MSIT
Capstone Project and Research 1 Instructor

Recommending Approval:


MR. MARVIN D.G. GARCIA, MSIT
CICT Chairperson

Conforme:


MRS. MARY ANN L. DE GUZMAN
IT Expert



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

October 3, 2025

MR. MARK CEDRICK NAGAÑO
IT Manager
Nitos Enterprises Corporation
San Anton, San Leonardo, Nueva Ecija

Sir,

Greetings!

We, the Bachelor of Science in Information Technology students, are in the final stage of stages of writing our Capstone Project entitled “**Barangay-Connect: Barangay Household Profiling System ++ for San Isidro, Nueva Ecija**”

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Respectfully yours,

ART D. ANGELES

RJ G. SOTTO

RICKY JR. P. DELA CRUZ

AVEGAIL V. PANGILINAN

LEMUEL M. REBULTAN

Noted:

Recommending Approval:

MRS. ARLIE C. SANTOS, MSIT
Capstone Project and Research 1 Instructor

MR. MARVIN D.G. GARCIA, MSIT
CICT Chairperson

Conforme:

MR. MARK CEDRICK NAGAÑO
IT Expert



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

October 3, 2025

MR. MARK CEDRICK NAGAÑO
IT Manager
Nitos Enterprises Corporation
San Anton, San Leonardo, Nueva Ecija

Sir,

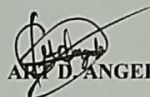
Greetings!

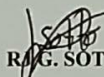
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Respectfully yours,


ARNEL ANGELES

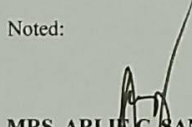

R.G. SOTTO


RICKY JR. P. DELA CRUZ


AVEGAIL V. PANGILINAN


LEMUEL M. REBULTAN

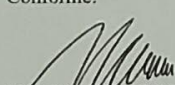
Noted:


MRS. ARLIE C. SANTOS, MSIT
Capstone Project and Research 1 Instructor

Recommending Approval:


MR. MARVIN D.G. GARCIA, MSIT
CICT Chairperson

Conforme:


MR. MARK CEDRICK NAGAÑO
IT Expert



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

October 4, 2025

MR. NIKKO ESTEBAN
Frontend Developer
Vtimetech Consulting, Inc.

Sir,

Greetings!

We, the Bachelor of Science in Information Technology students, are in the final stage of stages of writing our Capstone Project entitled “**Barangay-Connect: Barangay Household Profiling System ++ for San Isidro, Nueva Ecija**”

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RJ G. SOTTO

RICKY JR. P. DELA CRUZ

AVEGAIL V. PANGILINAN

LEMUEL M. REBULTAN

Noted:

Recommending Approval:

MRS. ARLIE C. SANTOS, MSIT
Capstone Project and Research I Instructor

MR. MARVIN D.G. GARCIA, MSIT
CICT Chairperson

Conforme:


MR. NIKKO ESTEBAN
IT Expert



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

October 4, 2025

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Vtimetech Consulting, Inc.

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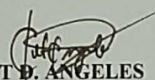
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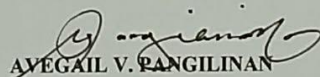
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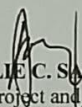

LEMUEL M. REBULTAN


ARLIE C. SANTOS, MSIT
Capstone Project and Research 1 Instructor


MARVIN D.G. GARCIA, MSIT
CICT Chairperson

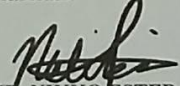
Noted:

Recommending Approval:


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Capstone Project and Research 1 Instructor


MR. MARVIN D.G. GARCIA, MSIT
CICT Chairperson

Conforme:


MR. NIKKO ESTEBAN
IT Expert



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

October 5, 2025

MR. MARK MESINA REYES
Junior Software Developer
Mira Software Solutions
Poblacion, San Isidro, Nueva Ecija

Sir,

Greetings!

We, the Bachelor of Science in Information Technology students, are in the final stage of stages of writing our Capstone Project entitled “**Barangay-Connect: Barangay Household Profiling System ++ for San Isidro, Nueva Ecija**”

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AVEGAIL V. PANGILINAN

LEMUEL M. REBULTAN

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Capstone Project and Research 1 Instructor

MR. MARVIN D.G. GARCIA, MSIT
CICT Chairperson

Conforme:

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NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

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MRS. ARLENE C. SANTOS, MSIT
Capstone Project and Research I Instructor

Recommending Approval:

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CICT Chairperson

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IT Expert



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SAN ISIDRO CAMPUS – College of Information and Communications Technology

October 5, 2025

MR. ALDRIN F. MANANSALA
Teacher II
Cabiao Senior National High School
Cabiao Nueva Ecija

Sir,

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Capstone Project and Research 1 Instructor

MR. MARVIN D.G. GARCIA, MSIT
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Conforme:

MR. ALDRIN F. MANANSALA
IT Expert



Republic of the Philippines

NUEVA ECIIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

September 8, 2025

ALDRIN F. MANANSALA

Teacher II

Cabiao Senior National High School

Cabiao

Nueva Ecija

Sir,

Greetings!

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R. G. SOTTO

RICKY JR. P. DELA CRUZ

AVEGAIL V. PANGILINAN

LEMUEL M. REBULTAN

Noted:

Recommending Approval:

MRS. ARCHIE C. SANTOS, MSIT
Capstone Project and Research 1 Instructor

MR. MARVIN D.G. GARCIA, MSIT
CICT Chairperson

Conforme:

MR. ALDRIN F. MANANSALA
IT Expert

APPENDIX D
Personal Information of IT Experts



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

IT Expert

Personal Profile

Name: ALDRIN F. MANANSALA Gender: MALE

Company: DEPED

Department: TVL-ICT

Project Developed: Auto Supply Inventory System

Job Description: Teacher II Civil Status: Married

Age: 37

Number of Years in Practice: 15 years

Educational Attainment

	Course	Year Graduated/Units Taken	School
Bachelors Degree	BSIT	2009	NEUST
Master Degree	MSIT	36 UNITS	NEUST
Doctorial Degree			

Others, please specify:



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

IT Expert

Personal Profile

Name: NIKKO ESTEBAN Gender: MALE

Company: VTIMETECH CONSULTING, INC.

Department: WEB DEVELOPMENT

Project Developed: SENIOR CITIZEN SYSTEM

Job Description: Teacher II Civil Status: SINGLE

Age: 23 Number of Years in Practice: 3 years

Educational Attainment

	Course	Year Graduated/Units Taken	School
Bachelors Degree	BSIT	4	POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
Master Degree			
Doctorial Degree			

Others, please specify:



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

IT Expert

Personal Profile

Name: MARK CEDRIK NAGANO Gender: MALE

Company: NITOS ENTERPRISES CORPORATION

Department: IT DEPARTMENT

Project Developed: _____

Job Description: IT MANAGER Civil Status: SINGLE

Age: 30 Number of Years in Practice: 7 years

Educational Attainment

	Course	Year Graduated/Units Taken	School
Bachelors Degree	BSIT	2018	NEUST
Master Degree			
Doctorial Degree			

Others, please specify:



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

IT Expert

Personal Profile

Name: MARK REYES Gender: MALE

Company: MIRA SOFTWARE SYSTEMS

Department: LT DEPARTMENT

Project Developed: COMPUTER LABORATORY ATTENDANCE MONITORING SYSTEM

Job Description: JUNIOR SOFTWARE DEVELOPER Civil Status: SINGLE

Age: 23 Number of Years in Practice: 1 year

Educational Attainment

	Course	Year Graduated/Units Taken	School
Bachelors Degree	BSIT	2024	NEUST
Master Degree			
Doctorial Degree			

Others, please specify:



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

IT Expert

Personal Profile

Name: MARY ANN L. DE GUZMAN Gender: FEMALE

Company: T.A. DIONISIO NATIONAL HIGH SCHOOL

Department: JUNIOR HIGH SCHOOL

Project Developed: _____

Job Description: Teacher III Civil Status: Married

Age: 33 Number of Years in Practice: 12 years

Educational Attainment

	Course	Year Graduated/Units Taken	School
Bachelors Degree	BACHELOR OF SECONDARY MAJOR IN COMPUTER EDUCATION	2012	BULACAN STATE UNIVERSITY
Master Degree	MASTER OF ARTS IN EDUCATION MAJOR IN EDUCATIONAL MNGT.	2019	COLLEGE OF IMMACULATE CONCEPTION
Doctorial Degree			

Others, please specify:

APPENDIX E

Evaluation Sheets for IT Experts

Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija

Evaluation Form

(For IT Expert)

Name (Optional):

Status- IT Expert

Direction: Please evaluate the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija based on the characteristics provided by ISO 25010. Use the scale described below. Check the number that correspond your answer.

Scale	DESCRIPTOR
5	There is a very strong evidence on the existence of the quality.
4	There is a strong evidence on the existence of the quality.
3	There is a essential evidence on the existence of the quality.
2	There are weak evidence on the existence of the quality.
1	There is no evidence on the existence of the quality.

Scoring Mode:

5- Very Functional

4- Functional

3- Needs Improvement

2- Poor

1- Very Poor

Criteria	Parameters	Descriptors	5	4	3	2	1
FUNCTIONABILITY SUSTAINABILITY Refers to the characteristics that represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions	Functional Completeness	The system covers all the specified tasks and objectives: - The developed system can create an admin account. - The developed system can view the resident profiles and information. - The developed system can print the resident information needed. - The developed system can set schedule of announcement. - The developed system can manage the user information. - The system can add, edit, and remove resident information and profiles. - The developed system can search the information need.		✓			
	Functional Correctness	The system shows the accuracy in results and inputs processed.		✓			
	Functional Appropriateness	The system's functions facilitate the accomplishment of specified tasks and objectives.	✓				

Scoring Mode:							
5- Very Efficient		2- Poor					
4- Efficient		1- Very Poor					
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
PERFORMANCE EFFICIENCY Refers to the performance relative to the amount of resources used under stated conditions.	Time-behavior	The system provides quick response time for a given output.		✓			
	Resource utilization	The system does not consume much in memory, CPU, disk and network usage.			✓		

Scoring Mode:							
5- Very Compatible			2- Poor				
4- Compatible			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
COMPATABILITY Refers to the degree to which a product, system or component can exchange information with other products, systems or components, and/or perform its required functions, while sharing the same hardware or software environment.	Co-existence	The system can perform its required functions efficiently, while sharing common environment and resources with other products, without detrimental impact on any other product.	✓				
	Interoperability	The system can exchange information and use the information that has been exchanged.	✓				

Scoring Mode:							
5- Very Usable			2- Poor				
4- Usable			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
USABILITY Refers to the degree to which a product or system can be used by specified users to achieved specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	Appropriateness recognizability	The system can be recognized by the users as to appropriateness of their needs.	✓				
	Learnability	The system can be learned by novice, expert and casual users.	✓				
	Operability	The system has attributes that make it easy to operate and control	✓				
	User error protection	The system protects users against making errors.			✓		
	User interface aesthetics	User interface enables pleasing and satisfying interaction to the user.			✓		

Scoring Mode: 5- Very Reliable 4- Reliable 3- Needs Improvement 2- Poor 1- Very Poor							
Criteria	Parameters	Descriptors	5	4	3	2	1
RELIABILITY Refers to the degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	Maturity	The system generate consistent result under normal operation.		✓			
	Availability	The system is operational when required for use.		✓			
	Recoverability	The system can recover the data directly affected and re-established the desired state of the system.		✓			

Scoring Mode: 5- Very Secured 4- Secured 3- Needs Improvement 2- Poor 1- Very Poor							
Criteria	Parameters	Descriptors	5	4	3	2	1
SECURITY refers to the degree to which a system protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization.	Confidentiality	The system ensures that data are accessible only to those authorized to have access.			✓		
	Integrity	The system prevents unauthorized access to, or modification of, computer programs or data		✓			

Scoring Mode:							
5- Very Maintainable							
4- Maintainable							
3- Needs Improvement							
2- Poor							
1- Very Poor							
Criteria	Parameters	Descriptors	5	4	3	2	1
MAINTAINABILITY Refers to the degree of effectiveness and efficiency with which a product or system can be modified to improve, correct or adapt it to changes in environment, and in requirements.	Modularity	The system is composed of discrete components such that a change to one component has minimal impact on other components.				✓	
	Reusability	An asset can be used in more than one system, or in building other assets.		✓			
	Analyzability	The system has ability to identify the root cause of failure within the system.		✓			
	Modifiability	The system can be effectively and efficiently modified without introducing defects or degrading existing product quality	✓				
	Testability	The system has less effort needed to verify (test) system changes.		✓			

Scoring Mode:							
5- Very Portable				2- Poor			
4- Portable				1- Very Poor			
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
PORTABILITY Refers to the degree of effectiveness and efficiency with which a system, product or component can be transferred from one hardware, software or other operational or usage environment to another.	Adaptability	The system can effectively and efficiently be adapted for different or evolving hardware, software, or other operational or usage environment.	✓				
	Installability	The system can be successfully installed and/or uninstalled in a specified environment.	✓				
	Replaceability	The system can replace another specified software product for the same purpose in the same environment.	✓				

Kindly give your comments and suggestions on the system as a whole.

1. Single Source of User Information
 - o User data should come from one source to avoid duplication and inconsistency.
 - o Suggestion: Allow the admin to add new users only under the Residents module. Then, in the Household module, let the admin select from the existing residents instead of creating new entries.
 - o Benefit: Any updates made to a resident's details will automatically reflect in their associated household, ensuring consistency and accuracy of data.
2. Enhance Announcement Details in the Activity Tab
 - o Add more input fields when creating an announcement, such as target user groups (e.g., Persons with Disabilities (PWD), minors below 18, senior citizens, etc.).
 - o Benefit: This makes announcements more specific and relevant, improving communication with the intended audience.
3. Resident User Dashboard
 - o In the user dashboard, residents should only see their own data and not the data of other residents.
 - o Suggestion: Provide an option for residents to update or request changes to their own information (with admin approval if necessary).
 - o Benefit: Improves privacy and gives residents more control over keeping their information accurate and up to date.

Thank you!

Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija

Evaluation Form

(For IT Expert)

Name (Optional):

Status- IT Expert

Direction: Please evaluate the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija based on the characteristics provided by ISO 25010. Use the scale described below. Check the number that correspond your answer.

Scale	DESCRIPTOR
5	There is a very strong evidence on the existence of the quality.
4	There is a strong evidence on the existence of the quality.
3	There is a essential evidence on the existence of the quality.
2	There are weak evidence on the existence of the quality.
1	There is no evidence on the existence of the quality.

Scoring Mode:

5- Very Functional

4- Functional

3- Needs Improvement

2- Poor

1- Very Poor

Criteria	Parameters	Descriptors	5	4	3	2	1
FUNCTIONABILITY SUSTAINABILITY Refers to the characteristics that represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions	Functional Completeness	The system covers all the specified tasks and objectives: - The developed system can create an admin account. - The developed system can view the resident profiles and information. - The developed system can print the resident information needed. - The developed system can set schedule of announcement. - The developed system can manage the user information. - The system can add, edit, and remove resident information and profiles. - The developed system can search the information need.	/				
	Functional Correctness	The system shows the accuracy in results and inputs processed.	/				
	Functional Appropriateness	The system's functions facilitate the accomplishment of specified tasks and objectives.	/				

Scoring Mode:							
5- Very Efficient			2- Poor				
4- Efficient			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
PERFORMANCE EFFICIENCY Refers to the performance relative to the amount of resources used under stated conditions	Time-behavior	The system provides quick response time for a given output.	/				
	Resource utilization	The system does not consume much in memory, CPU, disk and network usage.		/			

Scoring Mode:							
5- Very Compatible				2- Poor			
4- Compatible				1- Very Poor			
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
COMPATABILITY Refers to the degree to which a product, system or component can exchange information with other products, systems or components, and/or perform its required functions, while sharing the same hardware or software environment.	Co-existence	The system can perform its required functions efficiently, while sharing common environment and resources with other products, without detrimental impact on any other product.	/				
	Interoperability	The system can exchange information and use the information that has been exchanged.	/				

Scoring Mode:							
5- Very Usable							
4- Usable							
3- Needs Improvement							
2- Poor							
1- Very Poor							
Criteria	Parameters	Descriptors	5	4	3	2	1
USABILITY Refers to the degree to which a product or system can be used by specified users to achieved specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	Appropriateness recognizability	The system can be recognized by the users as to appropriateness of their needs.	/				
	Learnability	The system can be learned by novice, expert and casual users.	/				
	Operability	The system has attributes that make it easy to operate and control	/				
	User error protection	The system protects users against making errors.		/			
	User interface aesthetics	User interface enables pleasing and satisfying interaction to the user.		/			

Scoring Mode:							
5- Very Reliable				2- Poor			
4- Reliable				1- Very Poor			
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
RELIABILITY Refers to the degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	Maturity	The system generate consistent result under normal operation.	/				
	Availability	The system is operational when required for use.	/				
	Recoverability	The system can recover the data directly affected and re-established the desired state of the system.		/			

Scoring Mode:							
5- Very Secured			2- Poor				
4- Secured			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
SECURITY refers to the degree to which a system protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization.	Confidentiality	The system ensures that data are accessible only to those authorized to have access.		/			
	Integrity	The system prevents unauthorized access to, or modification of, computer programs or data	/				

Scoring Mode:							
5- Very Maintainable		2- Poor					
4- Maintainable		1- Very Poor					
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
MAINTAINABILITY Refers to the degree of effectiveness and efficiency with which a product or system can be modified to improve, correct or adapt it to changes in environment, and in requirements.	Modularity	The system is composed of discrete components such that a change to one component has minimal impact on other components.	/				
	Reusability	An asset can be used in more than one system, or in building other assets.	/				
	Analyzability	The system has ability to identify the root cause of failure within the system.	/				
	Modifiability	The system can be effectively and efficiently modified without introducing defects or degrading existing product quality	/				
	Testability	The system has less effort needed to verify (test) system changes.	/				

Scoring Mode:							
5- Very Portable			2- Poor				
4- Portable			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
PORTABILITY Refers to the degree of effectiveness and efficiency with which a system, product or component can be transferred from one hardware, software or other operational or usage environment to another.	Adaptability	The system can effectively and efficiently be adapted for different or evolving hardware, software, or other operational or usage environment.	/				
	Installability	The system can be successfully installed and/or uninstalled in a specified environment.	/				
	Replaceability	The system can replace another specified software product for the same purpose in the same environment.	/				

Kindly give your comments and suggestions on the system as a whole.

Thank you!

Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija

Evaluation Form

(For IT Expert)

Name (Optional):

Status- IT Expert

Direction: Please evaluate the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija based on the characteristics provided by ISO 25010. Use the scale described below. Check the number that correspond your answer.

Scale	DESCRIPTOR
5	There is a very strong evidence on the existence of the quality.
4	There is a strong evidence on the existence of the quality.
3	There is a essential evidence on the existence of the quality.
2	There are weak evidence on the existence of the quality.
1	There is no evidence on the existence of the quality.

Scoring Mode:

5- Very Functional

4- Functional

3- Needs Improvement

2- Poor

1- Very Poor

Criteria	Parameters	Descriptors	5	4	3	2	1
FUNCTIONABILITY SUSTAINABILITY Refers to the characteristics that represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions	Functional Completeness	The system covers all the specified tasks and objectives: - The developed system can create an admin account. - The developed system can view the resident profiles and information. - The developed system can print the resident information needed. - The developed system can set schedule of announcement. - The developed system can manage the user information. - The system can add, edit, and remove resident information and profiles. - The developed system can search the information need.	/				
	Functional Correctness	The system shows the accuracy in results and inputs processed.	/				
	Functional Appropriateness	The system's functions facilitate the accomplishment of specified tasks and objectives.		/			

Scoring Mode:							
5- Very Efficient		2- Poor					
4- Efficient		1- Very Poor					
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
PERFORMANCE EFFICIENCY Refers to the performance relative to the amount of resources used under stated conditions.	Time-behavior	The system provides quick response time for a given output.	/				
	Resource utilization	The system does not consume much in memory, CPU, disk and network usage.	/				

Scoring Mode: 5- Very Compatible 4- Compatible 3- Needs Improvement 2- Poor 1- Very Poor							
Criteria	Parameters	Descriptors	5	4	3	2	1
COMPATABILITY Refers to the degree to which a product, system or component can exchange information with other products, systems or components, and/or perform its required functions, while sharing the same hardware or software environment.	Co-existence	The system can perform its required functions efficiently, while sharing common environment and resources with other products, without detrimental impact on any other product.	/				
	Interoperability	The system can exchange information and use the information that has been exchanged.	/				

Scoring Mode:							
5- Very Usable		2- Poor					
4- Usable		1- Very Poor					
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
USABILITY Refers to the degree to which a product or system can be used by specified users to achieved specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	Appropriateness recognizability	The system can be recognized by the users as to appropriateness of their needs.	/				
	Learnability	The system can be learned by novice, expert and casual users.	/				
	Operability	The system has attributes that make it easy to operate and control	/				
	User error protection	The system protects users against making errors.		/			
	User interface aesthetics	User interface enables pleasing and satisfying interaction to the user.	/				

Scoring Mode:							
5- Very Reliable			2- Poor				
4- Reliable			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
RELIABILITY Refers to the degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	Maturity	The system generate consistent result under normal operation.	/				
	Availability	The system is operational when required for use.	/				
	Recoverability	The system can recover the data directly affected and re-established the desired state of the system.	/				

Scoring Mode:							
5- Very Secured				2- Poor			
4- Secured				1- Very Poor			
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
SECURITY refers to the degree to which a system protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization.	Confidentiality	The system ensures that data are accessible only to those authorized to have access.		/			
	Integrity	The system prevents unauthorized access to, or modification of, computer programs or data	/				

Scoring Mode:							
5- Very Maintainable		2- Poor					
4- Maintainable		1- Very Poor					
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
MAINTAINABILITY Refers to the degree of effectiveness and efficiency with which a product or system can be modified to improve, correct or adapt it to changes in environment, and in requirements.	Modularity	The system is composed of discrete components such that a change to one component has minimal impact on other components.	/				
	Reusability	An asset can be used in more than one system, or in building other assets.	/				
	Analyzability	The system has ability to identify the root cause of failure within the system.	/				
	Modifiability	The system can be effectively and efficiently modified without introducing defects or degrading existing product quality	/				
	Testability	The system has less effort needed to verify (test) system changes.	/				

Scoring Mode:							
5- Very Portable			2- Poor				
4- Portable			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
PORTABILITY Refers to the degree of effectiveness and efficiency with which a system, product or component can be transferred from one hardware, software or other operational or usage environment to another.	Adaptability	The system can effectively and efficiently be adapted for different or evolving hardware, software, or other operational or usage environment.	/				
	Installability	The system can be successfully installed and/or uninstalled in a specified environment.	/				
	Replaceability	The system can replace another specified software product for the same purpose in the same environment.	/				

Kindly give your comments and suggestions on the system as a whole.

Thank you!

Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija

Evaluation Form

(For IT Expert)

Name (Optional):

Status- IT Expert

Direction: Please evaluate the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija based on the characteristics provided by ISO 25010. Use the scale described below. Check the number that correspond your answer.

Scale	DESCRIPTOR
5	There is a very strong evidence on the existence of the quality.
4	There is a strong evidence on the existence of the quality.
3	There is a essential evidence on the existence of the quality.
2	There are weak evidence on the existence of the quality.
1	There is no evidence on the existence of the quality.

Scoring Mode:							
5- Very Functional			2- Poor				
4- Functional			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
FUNCTIONABILITY SUSTAINABILITY Refers to the characteristics that represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions	Functional Completeness	The system covers all the specified tasks and objectives: - The developed system can create an admin account. - The developed system can view the resident profiles and information. - The developed system can print the resident information needed. - The developed system can set schedule of announcement. - The developed system can manage the user information. - The system can add, edit, and remove resident information and profiles. - The developed system can search the information need.	/				
	Functional Correctness	The system shows the accuracy in results and inputs processed.	/				
	Functional Appropriateness	The system's functions facilitate the accomplishment of specified tasks and objectives.	/				

Scoring Mode:							
5- Very Efficient		2- Poor					
4- Efficient		1- Very Poor					
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
PERFORMANCE EFFICIENCY Refers to the performance relative to the amount of resources used under stated conditions.	Time-behavior	The system provides quick response time for a given output.	/				
	Resource utilization	The system does not consume much in memory, CPU, disk and network usage.		/			

Scoring Mode: 5- Very Compatible 4- Compatible 3- Needs Improvement							2- Poor 1- Very Poor	
Criteria	Parameters	Descriptors	5	4	3	2	1	
COMPATABILITY Refers to the degree to which a product, system or component can exchange information with other products, systems or components, and/or perform its required functions, while sharing the same hardware or software environment.	Co-existence	The system can perform its required functions efficiently, while sharing common environment and resources with other products, without detrimental impact on any other product.	/					
	Interoperability	The system can exchange information and use the information that has been exchanged.	/					

Scoring Mode:							
5- Very Usable			2- Poor				
4- Usable			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
USABILITY Refers to the degree to which a product or system can be used by specified users to achieved specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	Appropriateness recognizability	The system can be recognized by the users as to appropriateness of their needs.	/				
	Learnability	The system can be learned by novice, expert and casual users.	/				
	Operability	The system has attributes that make it easy to operate and control	/				
	User error protection	The system protects users against making errors.		/			
	User interface aesthetics	User interface enables pleasing and satisfying interaction to the user.		/			

Scoring Mode:							
5- Very Reliable			2- Poor				
4- Reliable			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
RELIABILITY Refers to the degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	Maturity	The system generate consistent result under normal operation.	/				
	Availability	The system is operational when required for use.	/				
	Recoverability	The system can recover the data directly affected and re-established the desired state of the system.		/			

Scoring Mode:							
5- Very Secured				2- Poor			
4- Secured				1- Very Poor			
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
SECURITY refers to the degree to which a system protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization.	Confidentiality	The system ensures that data are accessible only to those authorized to have access.		/			
	Integrity	The system prevents unauthorized access to, or modification of, computer programs or data	/				

Scoring Mode:							
5- Very Maintainable		2- Poor					
4- Maintainable		1- Very Poor					
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
MAINTAINABILITY Refers to the degree of effectiveness and efficiency with which a product or system can be modified to improve, correct or adapt it to changes in environment, and in requirements.	Modularity	The system is composed of discrete components such that a change to one component has minimal impact on other components.	/				
	Reusability	An asset can be used in more than one system, or in building other assets.	/				
	Analyzability	The system has ability to identify the root cause of failure within the system.	/				
	Modifiability	The system can be effectively and efficiently modified without introducing defects or degrading existing product quality	/				
	Testability	The system has less effort needed to verify (test) system changes.	/				

Scoring Mode:							
5- Very Portable			2- Poor				
4- Portable			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
PORTABILITY Refers to the degree of effectiveness and efficiency with which a system, product or component can be transferred from one hardware, software or other operational or usage environment to another.	Adaptability	The system can effectively and efficiently be adapted for different or evolving hardware, software, or other operational or usage environment.	/				
	Installability	The system can be successfully installed and/or uninstalled in a specified environment.	/				
	Replaceability	The system can replace another specified software product for the same purpose in the same environment.	/				

Kindly give your comments and suggestions on the system as a whole.

Thank you!

Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija

Evaluation Form

(For IT Expert)

Name (Optional):

Status- IT Expert

Direction: Please evaluate the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija based on the characteristics provided by ISO 25010. Use the scale described below. Check the number that correspond your answer.

Scale	DESCRIPTOR
5	There is a very strong evidence on the existence of the quality.
4	There is a strong evidence on the existence of the quality.
3	There is a essential evidence on the existence of the quality.
2	There are weak evidence on the existence of the quality.
1	There is no evidence on the existence of the quality.

Scoring Mode: 5- Very Functional 4- Functional 3- Needs Improvement		2- Poor 1- Very Poor					
Criteria	Parameters	Descriptors	5	4	3	2	1
FUNCTIONABILITY SUSTAINABILITY Refers to the characteristics that represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions	Functional Completeness	The system covers all the specified tasks and objectives: - The developed system can create an admin account. - The developed system can view the resident profiles and information. - The developed system can print the resident information needed. - The developed system can set schedule of announcement. - The developed system can manage the user information. - The system can add, edit, and remove resident information and profiles. - The developed system can search the information need.		/			
	Functional Correctness	The system shows the accuracy in results and inputs processed.				/	
	Functional Appropriateness	The system's functions facilitate the accomplishment of specified tasks and objectives.			/		

Scoring Mode:							
5- Very Efficient			2- Poor				
4- Efficient			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
PERFORMANCE EFFICIENCY Refers to the performance relative to the amount of resources used under stated conditions.	Time-behavior	The system provides quick response time for a given output.		/			
	Resource utilization	The system does not consume much in memory, CPU, disk and network usage.		/			

Scoring Mode: 5- Very Compatible 4- Compatible 3- Needs Improvement							2- Poor 1- Very Poor	
Criteria	Parameters	Descriptors	5	4	3	2	1	
COMPATABILITY Refers to the degree to which a product, system or component can exchange information with other products, systems or components, and/or perform its required functions, while sharing the same hardware or software environment.	Co-existence	The system can perform its required functions efficiently, while sharing common environment and resources with other products, without detrimental impact on any other product.		/				
	Interoperability	The system can exchange information and use the information that has been exchanged.		/				

Scoring Mode:							
5- Very Usable		2- Poor					
4- Usable		1- Very Poor					
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
USABILITY Refers to the degree to which a product or system can be used by specified users to achieved specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	Appropriateness recognizability	The system can be recognized by the users as to appropriateness of their needs.		/			
	Learnability	The system can be learned by novice, expert and casual users.		/			
	Operability	The system has attributes that make it easy to operate and control	/				
	User error protection	The system protects users against making errors.			/		
	User interface aesthetics	User interface enables pleasing and satisfying interaction to the user.			/		

Scoring Mode:							
5- Very Reliable				2- Poor			
4- Reliable				1- Very Poor			
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
RELIABILITY Refers to the degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	Maturity	The system generate consistent result under normal operation.		/			
	Availability	The system is operational when required for use.		/			
	Recoverability	The system can recover the data directly affected and re-established the desired state of the system.		/			

Scoring Mode:							
5- Very Secured				2- Poor			
4- Secured				1- Very Poor			
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
SECURITY refers to the degree to which a system protects information and data so that persons or other products or systems have the degree of data access appropriate to their types and levels of authorization.	Confidentiality	The system ensures that data are accessible only to those authorized to have access.			/		
	Integrity	The system prevents unauthorized access to, or modification of, computer programs or data			/		

Scoring Mode:							
5- Very Maintainable		2- Poor					
4- Maintainable		1- Very Poor					
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
MAINTAINABILITY Refers to the degree of effectiveness and efficiency with which a product or system can be modified to improve, correct or adapt it to changes in environment, and in requirements.	Modularity	The system is composed of discrete components such that a change to one component has minimal impact on other components.		/			
	Reusability	An asset can be used in more than one system, or in building other assets.		/			
	Analyzability	The system has ability to identify the root cause of failure within the system.	/				
	Modifiability	The system can be effectively and efficiently modified without introducing defects or degrading existing product quality		/			
	Testability	The system has less effort needed to verify (test) system changes.		/			

Scoring Mode:							
5- Very Portable			2- Poor				
4- Portable			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
PORTABILITY Refers to the degree of effectiveness and efficiency with which a system, product or component can be transferred from one hardware, software or other operational or usage environment to another.	Adaptability	The system can effectively and efficiently be adapted for different or evolving hardware, software, or other operational or usage environment.			/		
	Installability	The system can be successfully installed and/or uninstalled in a specified environment.			/		
	Replaceability	The system can replace another specified software product for the same purpose in the same environment.			/		

Kindly give your comments and suggestions on the system as a whole.

Thank you!

APPENDIX F

Evaluation Sheets for End Users

Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija

Evaluation Form

(For End-Users)

Name (Optional):

Position Description: ☐ Secretary

Direction: Please evaluate the Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija based on the characteristics provided by ISO 25010. Use the scale described below. Check the number that correspond your answer.

Scale	DESCRIPTOR
5	There is a very strong evidence on the existence of the quality.
4	There is a strong evidence on the existence of the quality.
3	There is a essential evidence on the existence of the quality.
2	There are weak evidence on the existence of the quality.
1	There is no evidence on the existence of the quality.

Scoring Mode:

5- Very Functional

4- Functional

3- Needs Improvement

2- Poor

1- Very Poor

Criteria	Parameters	Descriptors	5	4	3	2	1
FUNCTIONABILITY SUSTAINABILITY Refers to the characteristics that represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions	Functional Completeness	The system covers all the specified tasks and objectives: - The developed system can create an admin account. - The developed system can view the resident profiles and information. - The developed system can print the resident information needed. - The developed system can set schedule of announcement. - The developed system can manage the user information. - The system can add, edit, and remove resident information and profiles. - The developed system can search the information need.					
	Functional Correctness	The system shows the accuracy in results and inputs processed.					
	Functional Appropriateness	The system's functions facilitate the accomplishment of specified tasks and objectives.					

Scoring Mode:							
5- Very Efficient		2- Poor					
4- Efficient		1- Very Poor					
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
PERFORMANCE EFFICIENCY Refers to the performance relative to the amount of resources used under stated conditions.	Time-behavior	The system provides quick response time for a given output.					
	Resource utilization	The system does not consume much in memory, CPU, disk and network usage.					

Scoring Mode:							
5- Very Usable			2- Poor				
4- Usable			1- Very Poor				
3- Needs Improvement							
Criteria	Parameters	Descriptors	5	4	3	2	1
USABILITY Refers to the degree to which a product or system can be used by specified users to achieved specified goals with effectiveness, efficiency, and satisfaction in a specified context of use,	Appropriateness recognizability	The system can be recognized by the users as to appropriateness of their needs.					
	Learnability	The system can be learned by novice, expert and casual users.					
	Operability	The system has attributes that make it easy to operate and control					
	User error protection	The system protects users against making errors.					
	User interface aesthetics	User interface enables pleasing and satisfying interaction to the user.					

Kindly give your comments and suggestions on the system as a whole.

Thank you!

APPENDIX G
Letter for Thesis Adviser

	Republic of the Philippines NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY Cabanatuan City, Nueva Ecija, Philippines ISO 9001:2015 CERTIFIED	
SAN ISIDRO CAMPUS – College of Information and Communications Technology		
September 4, 2025		
MRS. ELLEN JANE G. REYES Associate Professor I Nueva Ecija University of Science and Technology San Isidro Campus San Isidro, Nueva Ecija Ma'am, Greetings! We, the Bachelor of Science in Information Technology students, are in the preliminary stages of writing our Capstone Project entitled "Barangay Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija". It would be a great help for us if you will be part of our study as our Thesis Adviser. We believe that your expertise and technical knowledge will be a great help for us to fully accomplish the requirements of the course BSIT. We assure you that the time and guidance that you will share would be utilized responsibly. Your acceptance in this regard is highly appreciated. Respectfully yours,		
ART D. ANGELES	RJ G. SOTTO	RICKY JR. P. DELA CRUZ
AVEGAIL V. PANGILINAN		LEMUEL M. REBULTAN
Noted:		Conforme:
MRS. ARLIE C. SANTOS, MSIT Capstone Project and Research I Instructor		MRS. ELLEN JANE G. REYES Thesis Adviser
<i>Transforming Communities through Science and Technology</i>		Contact No. (0 44) 940-9980 Email: sanidro@neust.edu.ph www.neust.edu.ph

APPENDIX H
Letter for English Critic



Republic of the Philippines

NUEVA ECIJA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Cabanatuan City, Nueva Ecija, Philippines

ISO 9001:2015 CERTIFIED



SAN ISIDRO CAMPUS – College of Information and Communications Technology

October 22, 2025

MR. JESSIE T. GONZALES
Coordinator, International Linkages
NEUST-San Isidro Campus
Poblacion, San Isidro, Nueva Ecija

Sir

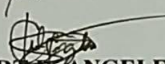
Greetings!

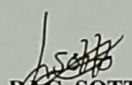
We, the Bachelor of Science in Informations Technology students, are in the preliminary stages of writing our Capstone Project entitled **“Barangay-Connect: Barangay Household Profiling System++ for San Isidro, Nueva Ecija”**.

It would be a great help for us if you will be part of our study as our **English Critic**. We believe that your technical knowledge in creative writing and English grammar will guide us in the proper approaches to complete our thesis. We assure you that the time and guidance that you will share would be utilized responsibly.

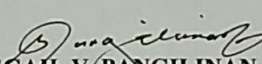
Your acceptance in this regard is highly appreciated.

Respectfully yours,


ART B. ANGELES


R. G. SOTTO


RICKY JR. P. DELA CRUZ

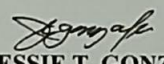

AVEGAIL V. PANGILINAN


LEMUEL M. REBULTAN

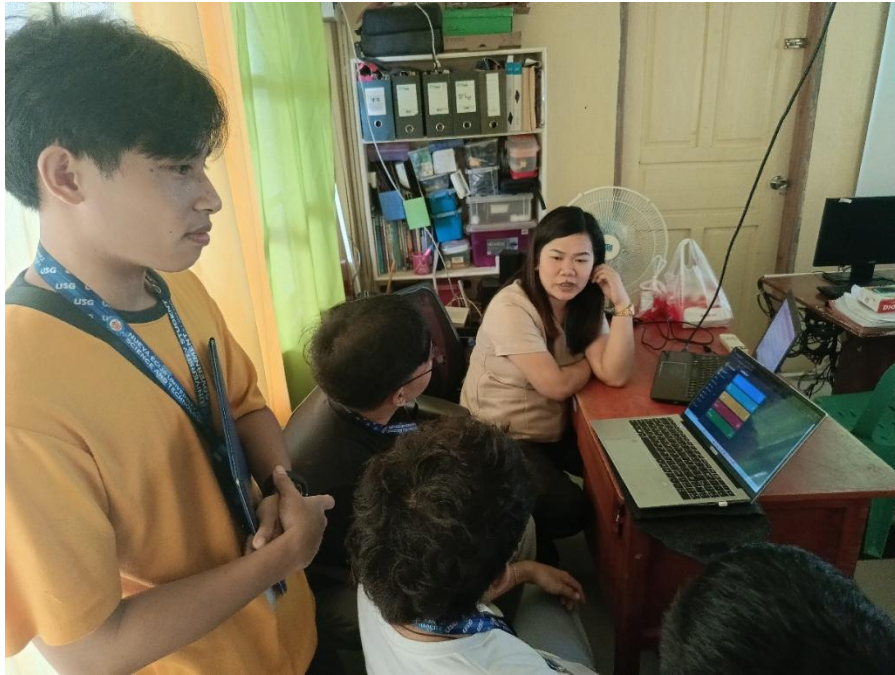
Noted:


MRS. ARLIE C. SANTOS, MSIT
Capstone Project and Research 1 Instructor

Conforme:


MR. JESSIE T. GONZALES
English Critic

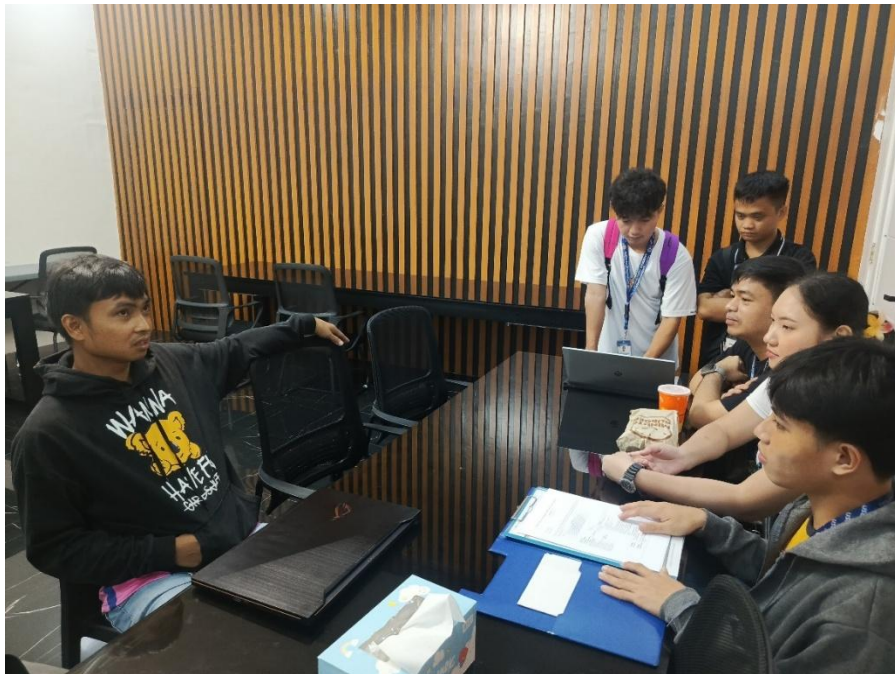
APPENDIX I
Actual IT Experts Validation Process



IT Experts Validation Process with Mrs. Mary Ann L. De Guzman on September 16, 2025 at T.A. Dionisio National High School



IT Experts Validation Process with Mr. Aldrin F. Manansala on September 09, 2025 at Cabiao National High School



IT Experts Validation Process with Mr. Mark Cedrick Nagaño on October, 2025 at Nitos Enterprise, San Leonard, Nueva Ecija

BSIT-4D Rebutan Lemuel M. (Presenting)

BSIT-4D Rebutan Lemuel M. Nikko Esteban BSIT-4D Rebutan Lemuel M. Ricky Jr. Dela Cruz

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People

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IN THE MEETING

Contributors 5

Ricky Jr. Dela Cruz (You) Meeting host

BSIT-4D Rebutan Lemuel M.

BSIT-4D Rebutan Lemuel M. Presentation

BSIT-4D Rebutan Lemuel M.

BSIT-4D Rebutan Lemuel M. (Presenting)

Barangay Household Management

ID	Last Name	First Name	Middle Name	Ext	Place of Birth	Birthdate	Age	Sex	Civil Status	Citizenship	Occupation	Indicatio	Household Size	Status	Actions
42	Ordo Mual	Sharon	Zahra	Martin	Spain	1991-05-24	33	Male	Widowed	In consanguinity MAJ	Est perferendis neq	PMO	5	Active	View
48	Maria Freeman	Sharon	Grady	Harmon	Mexico	1980-08-18	45	Female	Widowed	Blancas nobilitat	Est nuncupant nihil pl	Unemployed	1	Active	View
49	Karen Dignatus	Sharon	Wassie	Haggis	Spain	2012-05-15	12	Male	Single	Cuppa aptent vero i	Mauris veniam nra	Employed	1	Active	View
50	Osborn	Sharon	Clara	Sanchez	Spain	1980-11-28	44	Female	Widowed	Id vel sedem minima	Est qua conuenit lila	Unemployed	1	Active	View
51	Molten	Sharon	Francisco	Sanchez	Spain	2000-05-05	25	Male	Married	Et assumenda voluta	Nobis qui itaque a	Sole Parent	1	Active	View
52	Quamir Yalen	Sharon	Adrian	Dalton	Spain	1979-01-18	46	Female	Single	Vitaeque excepteur	Mollitia fugiat qui	Sole Parent	1	Active	View
53	Shed Ortega	Sharon	Blanca	Ward	Spain	2002-06-21	23	Male	Single	Non nuncupant indolita	Consequatur dolorem	OFW	1	Active	View
54	Claudia Shelton	Sharon	Casper	Salad	Spain	1979-10-07	46	Male	Single	Vitaeque nemo expe	Aute optent eu des	Unemployed	1	Active	View
55	Mia Guzman	Sharon	Clara	Gay	Spain	2007-08-25	18	Female	Single	Vitaeque qui i	Sit et laore off	Unemployed	1	Active	View
56	DELA CRUZ	Sharon	F		BAZONGA	2025-09-04	0	Male	Single	FILIPINO	NONE	Employed	1	Active	View

1:40 PM | czf-obzb-oti

Nikko Esteban

2 others

Ricky Jr. Dela Cruz

IT Experts Validation Process with Mr. Nikko Esteban on October 04, 2025 via Google Meet



IT Experts Validation Process with Mr. Mark Reyes on October 05, 2025 at Calaba, San Isidro, Nueva Ecija

ART DIONISIO ANGELES

SAN ISIDRO, NUEVA ECIJA

Mobile No: 09532579371

Email Address: angelesart28@gmail.com



OBJECTIVES:

Aspiring IT professional seeking a role in system administration and cyber security to support business operations and ensure secure technology solutions.

PERSONAL BACKGROUND

Age: 21

Date of Birth: February 27, 2004

Place of Birth: Cabanatuan City

Gender: Male

Height: 5'8

Weight: 69

Citizenship: Filipino

Civil Status: Single

Religion: Catholic

Mother's Name: Teresita Angeles

Occupation: Vendor

Father's Name: Luis Angeles

Occupation: Farmer

Guardian Contact No: 090829531393

EDUCATIONAL BACKGROUND

Tertiary	Nueva Ecija University of Science and Technology Bachelor of Science in Information Technology with Specialization in Web Systems Technology 2025 - Present
Senior High School	T.A. Dionisio National High School
Junior High School	T.A. Dionisio National High School
Elementary	San Isidro Elementary School

AWARDS

Dean's Lister.

STRENGTHS/SKILLS

Knowledge in programming fundamentals (e.g., Java, C++, python)

Team-oriented

Independent

RICKY JR P. DELA CRUZ

MALAPIT, SAN ISIDRO, NUEVA ECIJA

Mobile No: 09927683105

Email Address: sairidpineda28@gmail.com



OBJECTIVES:

IT graduate eager to support system maintenance, data management, and user assistance while enhancing my skills in modern technology.

PERSONAL BACKGROUND

Age: 21

Date of Birth: June 29, 2004

Place of Birth: Muntinlupa City

Gender: Male

Height: 5'5

Weight: 57

Citizenship: Filipino

Civil Status: Single

Religion: Catholic

Mother's Name: Lorie P. Dela Cruz

Occupation: housekeeper

Father's Name: Ricky Z. Dela Cruz

Occupation: Driver

Guardian Contact No: 09977808327

EDUCATIONAL BACKGROUND

Tertiary	Nueva Ecija University of Science and Technology Bachelor of Science in Information Technology with Specialization in Web Systems Technology 2025 - Present
Senior High School	T.A. Dionisio National High School
Junior High School	Ciriaco L. Icmantina Sr. National High school
Elementary	Bulabud Elementary school

AWARDS

Grade 8-With Honors

Grade 9-With Honors

Grade 10-With Honors

STRENGTHS/SKILLS:

Problem solving skills

Adaptability

Team Collaboration

Analytical thinking

Time management

Self learning

Knowledge in programming fundamentals (HTML, CSS, JAVA)

LEMUEL M. REBULTAN

PUROK 10, SAN VICENTE, CABIAO, NUEVA ECIJA

Mobile No: 09366337285

Email Address: rebultanlemuel@gmail.com



OBJECTIVES

I set to improve my skills and knowledge in Technology. These objectives guide my development so I can succeed and grow in the IT industry.

PERSONAL BACKGROUND

Age: 23

Date of Birth: July 8, 2002

Place of Birth: Cainta, Rizal

Gender: Male

Height: 5'1

Weight: 51

Citizenship: Filipino

Civil Status: Single

Religion: Catholic

Mother's Name: Leticia Rebultan

Occupation: housewife

Father's Name: Manuel Rebultan

Occupation: Operator

Guardian Contact No: 09366337285

EDUCATIONAL BACKGROUND

Tertiary	Nueva Ecija University of Science and Technology Bachelor of Science in Information Technology with Specialization in Web Systems Technology 2025 - Present
Senior High School	Cabiao National High School
Junior High School	Cabiao National High School
Elementary	San Vicente Elementary School

STRENGTHS/SKILLS

Teamwork

Collaboration

Independent

-My skill in IT is the ability to use technology effectively to solve problems, manage systems, and develop software or digital solutions.

RJ GREGORIO SOTTO

PUROK 1, SAN FERNANDO SUR, CABIAO, NUEVA ECIJA

Mobile No: 09489533298

Email Address: sotto.rj01@gmail.com

**OBJECTIVES**

To obtain a position as an IT Professional where i can apply my technical skills, problem-solving abilities, and knowledge in systems, networks, and web technologies to support organizational growth.

PERSONAL BACKGROUND

Age: 23

Date of Birth: October 14, 2002

Place of Birth: San Fernando, Sur, Cabiao, Nueva Ecija

Gender: Male

Height: 5'5

Weight: 51

Citizenship: Filipino

Civil Status: Single

Religion: Catholic

Mother's Name: Lourdes Sotto

Occupation: housewife

Father's Name: Rolando Sotto

Occupation: Fisherman

Guardian Contact No: 09489533298

EDUCATIONAL BACKGROUND

Tertiary	Nueva Ecija University of Science and Technology Bachelor of Science in Information Technology with Specialization in Web Systems Technology 2025 - Present
Senior High School	Cabiao National Senior High School
Junior High School	Cabiao National Junior High School
Elementary	San Fernando Sur Elementary School

ACHIEVEMENT BACKGROUND

Grade 11-With Honors

Grade 12-With Honors

STRENGTHS/SKILLS

Technical skills: Proficient in HTML, CSS, JavaScript, and PHP for front end and back-end web development.

Experience in developing a responsive and interactive web applications.

Database skills: Skilled in MySQL database design, management, and CRUD operations.

Soft Skills: Team work, problem solving, communication, time management and leadership.

AVEGAIL VELASCO PANGILINAN

PUROK 3, BRGY CASTELLANO, SAN LEONARDO, NUEVA ECIJA

Mobile No: 09755122949

Email Address: iamavegailpangilinan@gmail.com



OBJECTIVES

To successfully conclude my academic career, transition into a challenging and rewarding position at a top-tier industry leader, and actively commit to the continuous expansion of my professional skill set to drive organizational and personal growth.

PERSONAL BACKGROUND

Age: 21

Date of Birth: August 16, 2004

Place of Birth: Malapit, San Isidro, Nueva Ecija

Gender: Female

Height: 5'5

Weight: 70

Citizenship: Filipino

Civil Status: Single

Religion: Iglesia Ni Cristo

Mother's Name: Janet Pangilinan

Occupation: housewife

Father's Name: Femil Pangilinan

Occupation: Family Business

Guardian Contact No: 09362395189

EDUCATIONAL BACKGROUND:

Tertiary	Nueva Ecija University of Science and Technology Bachelor of Science in Information Technology with Specialization in Web Systems Technology 2025 - Present
Senior High School	Juan R. Liwag Memorial High School
Junior High School	Juan R. Liwag Memorial High School
Elementary	Castellano Elementary School

ACHIEVEMENT BACKGROUND

Grade 8-With Honors

Grade 9-With Honors

Grade 10-With Honors

STRENGTHS/SKILLS

Team Collaboration

Knowledge in programming fundamentals (HTML, CSS, JAVA)

Problem Solving